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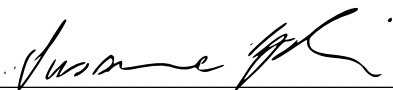
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Signature 

Typed name: Professor Markus Greiner, Chair

Signature 

Typed name: Professor Philip Kim

Signature 

Typed name: Professor Susanne Yelin

Date: February 4, 2021

A dipolar erbium quantum gas microscope

A DISSERTATION PRESENTED
BY
ANNE HÉBERT
TO
THE DEPARTMENT OF PHYSICS

IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF
DOCTOR OF PHILOSOPHY
IN THE SUBJECT OF
PHYSICS

HARVARD UNIVERSITY
CAMBRIDGE, MASSACHUSETTS
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A dipolar erbium quantum gas microscope

ABSTRACT

Quantum gas microscopy is an exciting platform for the study of *in situ* atom-atom interactions. Erbium, a dipolar atom, presents an opportunity to extend previous quantum gas microscopy experiments to more complex systems influenced by long-range, anisotropic interactions. In this thesis, I present progress towards a quantum gas microscope of erbium atoms, including details on the design of a novel accordion lattice system. This system will allow us to dynamically tune the spacing of both the vertical and horizontal optical lattices confining the atoms. In addition, I present on the sub-second generation of erbium Bose-Einstein condensates, made possible by implementing a two-stage magneto-optical trap and a crossed optical dipole trap with a tunable geometry.

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CETTE THÈSE EST POUR PAPA.

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1

Introduction

Quantum gas microscopy is a technique allowing us to control and image individual atoms in an optical lattice potential. Single-site addressability offers the possibility of imaging individual atoms *in situ*, as well as the ability of locally manipulating internal degrees of freedom and the potential landscapes. The first quantum gas microscopes were developed in the late 2000s [1], [2], building off of decades of expertise and advances in the atomic physics community. These two first microscopes used a bosonic isotope of rubidium, but over the next few years the field of quantum gas microscopy

spread to include many more elements. Around 2015, quantum gas microscopes using fermionic isotopes of lithium [3], [4], [5] and potassium [6], [7], [8] achieved single atom resolution, adding the possibility of studying systems with Fermi statistics. Further expanding the arsenal of quantum gas microscopy, a few groups developed single-site imaging of the lanthanide atom ytterbium [9], [10], also around 2015. The previous elements all being alkalis, this added the more complex electronic structure of a lanthanide to the possibilities of quantum gas microscopy.

Conceptually, a quantum gas microscope (QGM) is simply an ultracold atom machine that places a degenerate quantum gas of atoms in an optical lattice that is at the focus of a high-resolution imaging system. In order to be isolated from the environment, the atoms reside in a vacuum chamber, which makes it challenging to image them with the high numerical aperture required for single-site resolution. Depending on the specifics of each QGM design, the optics for the high numerical aperture imaging are either completely outside of the vacuum, partly inside and partly outside the vacuum, or, as in our design, the optics are completely confined to inside the vacuum chamber. Typically, in a QGM, the atoms are confined to one plane of a vertical periodic optical lattice, and are individually trapped in wells created by periodic potentials along the other two dimensions. In this way, the atoms reside in a two-dimensional crystal structure.

This lattice structure creates interesting model systems from condensed matter physics, in a clean controllable environment. This has allowed very successful and informative studies of antiferromagnetic spin chains [11], spin correlations and transport in the Hubbard model [12], [13], [14] and [15], thermalization and many-body localization [16], [17], [18] and [19], amongst many others.

A natural development of all of this promising work is to broaden even further the possible avenues of study of quantum gas microscopy. With this in mind, we decided to build a QGM using the lanthanide atom erbium, because of its large permanent magnetic dipole moment. This provides us with an additional interaction term: the dipolar interaction. Two dipolar erbium atoms residing on neighboring sites in the optical lattice will interact with each other via this dipole-dipole

interaction, which scales as $1/r^3$. This adds a long-range interaction to the on-site interactions and the tunneling.

The complex electronic structure of the lanthanides, which brings us the long-range dipolar interaction in some of the lanthanide atoms, also adds much more richness to the system. This electronic structure brings a variety of convenient atomic transitions with linewidths spanning from the MHz to the mHz. This allows us to implement techniques such as narrow-line magneto-optical trapping and optical molasses cooling in an optical dipole trap. These advances have allowed us to develop an extremely efficient sequence for both laser cooling and evaporatively cooling the erbium atoms, which has enabled us to reduce the BEC preparation time. Our results show a nearly pure BEC of 80,000 erbium atoms, prepared in a total of 850 ms. This is an enormous increase in speed relative to the standard in the field of quantum gas microscopy. The techniques used to achieve this sub-second BEC production, as well as the results, are detailed in this thesis.

Erbium was first trapped in a magneto-optical trap in 2006 by McClelland and Hanssen, at NIST [20]. Very shortly after, the first quantum degenerate gases of erbium were achieved by the Ferlaino group in Innsbruck, in 2012 [21] for the Bose-Einstein condensates, and in 2014 [22] for the degenerate Fermi gases. Since then, much has been learned by the ultracold atom community about working with dipolar atoms, and erbium in particular. First signatures of the dipolar interactions in erbium were observed in fermions in the interplay between the magnetic dipole-dipole interaction and the Pauli exclusion principle in [23]. In bosons, signatures of the dipole-dipole interaction were measured in realizing an extended Bose-Hubbard model, revealing the anisotropy of the interaction, as well as its long-range nature, in [24].

1.1 THESIS OUTLINE

This thesis is structured as follows:

- **Chapter 1:** Here I give a brief introduction to erbium and some of its properties. I also provide a brief overview of our apparatus, to give context for the work.
- **Chapter 2:** Here I describe the two magneto-optical traps used in our experiment, one using the 583 nm transition and the other using the 841 nm transition. I show data on the loading rates and lifetimes of each of the MOTs, as well as on the transfer between the two. Lastly, I show spectroscopy results measuring the isotope shift of the 841 nm transition.
- **Chapter 3:** Here I describe our experimental sequence for creating sub-second Bose-Einstein condensates of erbium. I give details on the optical setups and the parameters used in the sequence. I also discuss our degenerate Fermi gas results.
- **Chapter 4:** Here I show the magnetic field coils used in our experiment. I describe their designs and applications. I also give details on our winding methods, especially for the assembly of the MOT and Feshbach (offset) coils that is mounted on the main chamber of the experiment.
- **Chapter 5:** Here I describe the design of the accordion optical lattices. I describe the motivation for the vertical and horizontal accordion lattices, as well as the design constraints and how we overcame them. I give details on the design and construction, assembly and alignment of all the key components of this setup. I also include preliminary images showing the expansion of the lattice (with no atoms).

2

Background

2.1 ERBIUM

Erbium is a lanthanide, a rare earth element, with an atomic number of 68. Lanthanides first came of interest to the ultracold atomic physics community due to the many interesting properties brought by their complex electronic structures. Some of these, along with other relevant properties about erbium, will be discussed below.

Natural abundance of erbium isotopes			
Abundance (%)	Mass (amu)	Statistic	Nuclear spin
0.1	162	Bosonic	0
1.6	164	Bosonic	0
33.5	166	Bosonic	0
22.9	167	Fermionic	7/2
27.0	168	Bosonic	0
14.9	170	Bosonic	0

Table 2.1: Natural abundance and mass of the 6 stable isotopes of erbium, take from [25]. Also included are the nuclear spins and the statistics of each isotope.

2.1.1 ISOTOPES

Erbium has 6 naturally stable isotopes. Their relative natural abundances, their masses, their statistics and their nuclear spins are shown in Table 2.1. The data on natural abundances is taken from [25].

Theoretical predictions and a partial measurement of the isotope shifts in the relevant optical transitions of erbium were all found to be within a few GHz of the transition frequency for ^{166}Er [26]. This means that it is possible to design the laser systems such that they can be conveniently switched from operating on the transition frequency of one isotope to the transition frequency of another isotope. We completed a full set of measurements of the frequencies of the 401 nm, the 583 nm and the 841 nm transitions for all 6 isotopes, as shown in [27], confirming that they are all indeed within ± 3 GHz of the transition frequency for ^{166}Er .

This convenient isotope shift and the fact that erbium has stable bosonic and fermionic isotopes gives us the opportunity to build an apparatus with which we can easily switch from studying bosonic to fermionic statistics. This is extremely appealing, especially as no quantum gas microscope to date has this flexibility.

2.1.2 ELECTRON CONFIGURATION

The electron configuration of erbium is shown below, from [28]:

$$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^6 4f^{12} 6s^2$$

Note that the $4f$ orbitals are incomplete, having only 12 electrons where there is room for 14 electrons. Meanwhile the $6s$ orbitals accepts only 2 electrons, and are therefore full. The 2 “missing” electrons in the $4f$ orbitals while the $6s$ orbitals are full is called a submerged-shell configuration.

This complex configuration leads to a ground state orbital angular momentum of $L=5$, which is very large compared to most elements previously used in the cold-atom community. The bosonic isotopes have a nuclear spin of $I = 0$, which results in the bosonic isotopes not having a hyperfine structure. The fermionic isotope ^{167}Er has a nuclear spin of $I = 7/2$, leading to a hyperfine structure.

The unusual electron configuration is also responsible for erbium’s large permanent magnetic dipole moment, as detailed in the section below.

2.1.3 MAGNETIC DIPOLE MOMENT

Erbium has a large permanent atomic magnetic moment, of $\mu = 7\mu_B$, where μ_B is the Bohr magneton. Erbium is the element with the third highest magnetic moment, after dysprosium and terbium with $\mu = 10\mu_B$, and holmium with $\mu = 8\mu_B$.

Typical alkalis have $\mu = 1\mu_B$. Since the dipole-dipole interaction scales as the square of the dipole moment, this means that the dipole-dipole interaction between two erbium atoms is about 50 times stronger than the dipole-dipole interaction between two alkali atoms.

We can understand how the magnetic moment is a direct result of the electron configuration by

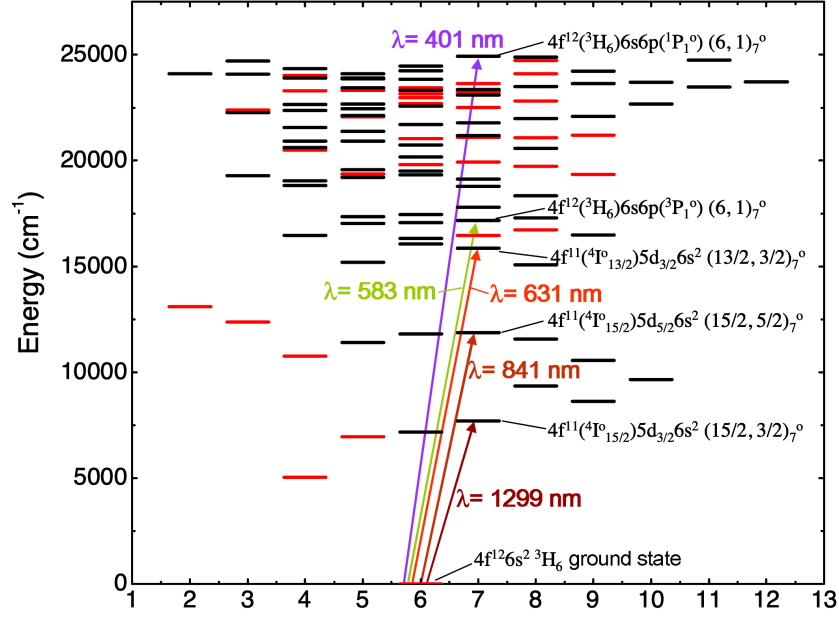


Figure 2.1: Energy levels of erbium. The five notable laser cooling transitions are labeled by their wavelengths. The red lines represent states with even parity and the black lines represent states with odd parity. The data and plot are from [29].

looking at the following expression for the magnetic moment:

$$\mu = m_J g_J \mu_B$$

The Landé g-factor for erbium is $g_J = 1.1638$, and m_J is the magnetic quantum number. In the stretched states, $m_J = \pm 6$, leading to dipole moment of $\mu = 6.98\mu_B$.

2.1.4 OPTICAL TRANSITIONS

A partial set of the energy levels of erbium are shown in Figure 2.1, from [29]. In our experiment, we are currently using the 401 nm, 583 nm and 841 nm transitions. Properties of these transitions are found in Table 2.2.

The range of linewidths of these optical transitions, from 30 MHz to 2 Hz, with conveniently

Erbium transitions	
Wavelength (nm)	Transition rate
401	27.5 MHz
583	190 kHz
631	28 kHz
841	8 kHz
1299	2.1 Hz

Table 2.2: Transition linewidths for five optical transitions in erbium, from [29].

spaced transitions in the 100-10 kHz range gives us a diverse set of tools with which to interact with the atoms.

2.2 OVERVIEW OF APPARATUS

Here I will include a brief overview of the apparatus, to provide context for the sections that are discussed in more detail in the subsequent chapters. The sections of the apparatus and the earlier portions of the experimental sequence that I am not describing in much detail are also covered in more depth in the two other theses from this experiment.

Our vacuum chamber is shown in Figure 2.2. The erbium atoms start in an effusion cell, as described in detail in [28] and [27]. The erbium atoms are in a tantalum crucible, through which a high current is run, raising the temperature of the atoms to 1100°C . This produces a vapor of erbium atoms. A set of small apertures selects the atoms traveling within a specific angle range, thus creating a beam of erbium atoms traveling down the vacuum chamber. The erbium used in the oven is the naturally-occurring mix of the 6 stable isotopes of erbium.

The first active interaction we have with these atoms is with a 2D molasses cooling section (as labeled on Figure 2.2) operating on the broad 401 nm transition, which slows the transverse velocity of the atoms and collimates the atom beam. The molasses was found to increase the flux into the

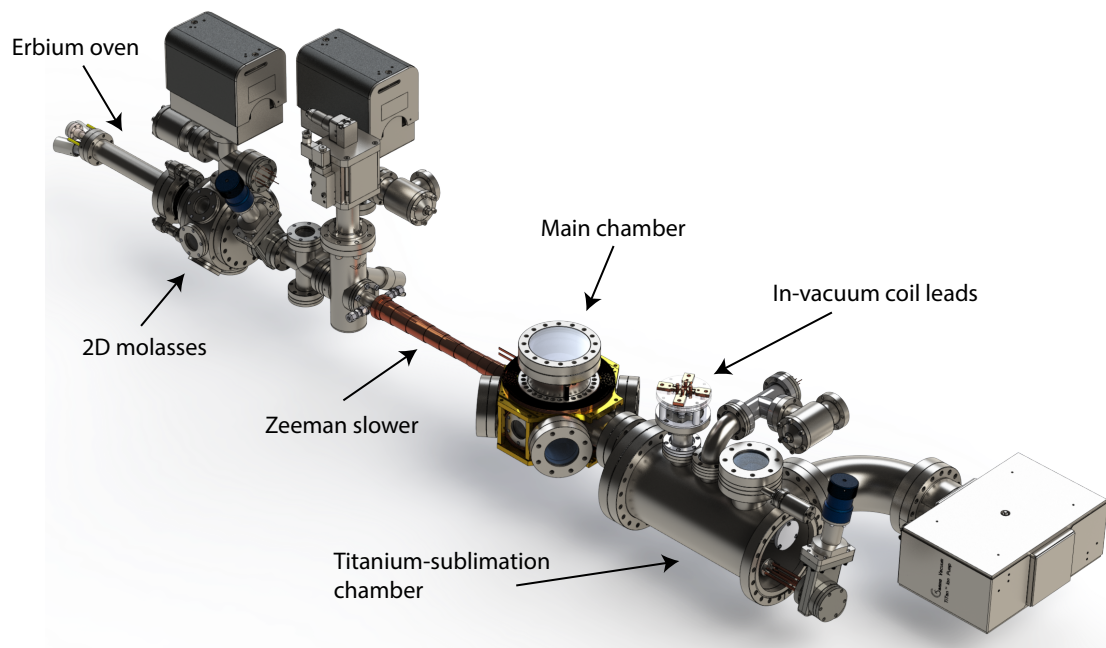


Figure 2.2: Rendering of the full vacuum chamber, showing the erbium oven, the 2D molasses section with its view-ports, the Zeeman slower tube, the main chamber and finally the titanium-sublimation chamber with the viewport through which the Zeeman slower beam is aligned.

main chamber, and increased the loading rate of the magneto-optical trap by a factor of more than 30. Part of the reason the molasses section is so effective is due to how broad the 401 nm transition is, allowing the atoms to scatter many photons even during the short interaction window. Additionally, the molasses beams are designed to be highly elliptical, in order to maximize the spatial overlap with the atom beam.

Next, the atoms are slowed axially by an increasing-field, spin-flip Zeeman slower, shown in Figure 2.2). The Zeeman slower coils are wound from flat, enameled copper wire, around a 360 mm long tube. The wire is coated with a heat-resistant insulation. The Zeeman slower tube has an outer layer allowing for water-cooling to run underneath the coils. For more details on the coils, see Section 5.3. In addition, see [27] and [28] for more details on the design of the Zeeman slower and the motivations behind the design choices. The Zeeman slower light is again operating on the 401 nm atomic transition. After a careful optimization of the currents in the different coils by measuring the velocity profile of the atoms, our Zeeman slower slows the atoms down to a velocity distribution peaked around 5-10 m/s.

With the atoms slowed down to 5-10 m/s, once they exit the Zeeman slower tube and enter the main chamber, they are now within the capture velocity of our first magneto-optical trap (MOT) operating on the 583 nm transition. The 583 nm MOT and the 841 nm MOT are described in detail in Chapter 3. The two-stage MOT, operating in the main chamber (labeled on Figure 2.2), gives us a method of rapidly creating very high phase-space density samples of atoms. In 400 ms, we can cool 4×10^7 atoms to about 0.5 μ K.

The main chamber, and the nearby vacuum components are designed to be as non-magnetic as possible. The main chamber is made from 316 stainless steel, an austenitic stainless steel. It was manufactured by Anderson Dahlen, Inc. The viewports on the main chamber are also made from the same steel, with a broadband anti-reflection coating. They were manufactured and coated for us by MPF Products, Inc.

From the second-stage 841 nm MOT, these atoms are then loaded into a tunable, crossed optical dipole trap, operating at 1064 nm. In the ODT, the atoms are optically cooled and then evaporatively cooled to degeneracy. As detailed in Chapter 4, with these methods we are able to produce nearly pure BECs of 80,000 atoms of ^{168}Er in 850 ms, as well as DFGs of ^{167}Er in about 5 s. This stage also happens in the main chamber, the same as the MOTs. In fact, the atoms spend the remainder of the experimental sequence in this main chamber.

The next step after reaching quantum degeneracy is then to load the atoms in the optical lattices. The lattices are discussed more in Chapter 4, as well as detailed discussion of the motivation and design for the accordion lattices.

As mentioned above, the optical lattices and the high-resolution imaging all take place in the main chamber. The high-resolution imaging will be done via a high-numerical aperture, custom microscope objective designed and assembled for us by Special Optics. Details on the design and testing of the objective, as well as the mounting in the vacuum chamber can be found in Aaron Krahn's thesis [30].

The high-resolution imaging will be fluorescence imaging done on the broad 401 nm transition. The choice of using the broad transition is motivated by how quickly the atoms can re-scatter photons on this transition. Indeed, with a 27.5 MHz linewidth, the atoms scatter about 10^8 photons/second. This means that the imaging time could be reduced to 1-2 microseconds, and the atoms would still have scattered hundreds of photons each. If the imaging system has high enough numerical aperture and has low enough photon losses, this is in principle a sufficient number of photons to resolve the individual atoms. Another important property of erbium that impacts the imaging scheme is its very large mass of 168 amu. This means that the photon recoil is less than with typical alkalis, which have recoil temperatures closer to $1\mu\text{K}$.

$$T_{\text{recoil}} = \frac{\hbar^2 k^2}{2mk_B} = 350\text{nK}$$

This means that while absorbing and re-scattering the 401 nm photons for imaging, the erbium atoms will experience minimal displacement, particularly if the imaging pulse is kept to a few microseconds. This opens the door to simpler imaging schemes, without needing to pin the atoms in a deep lattice during imaging, or resort to cooling during the imaging. Simulations showing the algorithms for identifying the positions of the atoms, and discussions of the EMCCDs can be found in [30] and [27].

3

841 nm MOT and spectroscopy

As the ultracold atom community has expanded its reach to include more elements outside of just the alkalis, narrow-line magneto-optical traps have become more and more widespread and appreciated. In the lanthanides, several groups have successfully demonstrated and used MOTs operating on the 190 kHz transition of erbium ([31] and [32] amongst others), the 130 kHz transition of dysprosium ([33] and [34] amongst others) and the 180 kHz transition of ytterbium (for example [35]). The narrow 7 kHz transition in strontium has also been used to operate the second stage of

a two-stage MOT, for example in [36]. We were inspired by the success of these two-stage MOTs to explore the possibilities of implementing similar techniques in erbium.

We operate a first MOT on the 583 nm transition which is 190 kHz broad, as in [31]. We then transfer the atoms into a second-stage MOT on the 841 nm transition, which is 8 kHz broad. While a blue-detuned MOT on this 841 nm transition of erbium was achieved in [37], to our knowledge, this is the first red-detuned erbium MOT operating on this transition.

In this chapter I describe the details of the operation of our two magneto-optical traps, the first on the 583 nm transition (which is 190 kHz wide), and then the hand-off to the second, narrow-line MOT operating on the 841 nm transition (which is 8 kHz wide). Lastly, I describe some spectroscopy measurements that we have done using our narrow-line MOTs.

3.1 583 NM MOT

Magneto-optical traps (MOTs) are an important laser-cooling technique which is commonly used in cold-atom experiments. In a MOT, the atoms can be cooled up to but not beyond the Doppler temperature, which is set by the linewidth of the atomic transition on which the MOT is operating. The Doppler temperature is given by $T_{\text{Doppler}} = \frac{\hbar\Gamma}{2k_B}$. For typical alkali experiments, such as rubidium, lithium, cesium etc, the Doppler temperature is in the 100 μK range. This is still several orders of magnitude away from the critical temperature at which quantum degeneracy is achieved, which is typically in the 10 – 100 nK range. For this reason, MOTs are typically followed in the experimental sequence by less efficient cooling methods such as evaporative cooling in an optical-dipole trap.

By operating a MOT on a narrower transition, with a lower Doppler temperature limit, we can then create colder atom clouds while still using only the more efficient laser cooling. By limiting the amount of cooling for which we rely on less efficient cooling methods, we can then make our whole experimental sequence more efficient.

In erbium we have several atomic transitions in a convenient regime for operating such narrow-line MOTs. For example, we have the 583 nm transition, which has a linewidth of $\Gamma_{583} = 2\pi \times 190$ kHz, with a Doppler temperature of $T_{\text{Doppler}} = \frac{\hbar\Gamma}{2k_B} = 4.8\mu\text{K}$. This is already almost two orders of magnitude colder than typical alkali MOTs. For even narrower lines, erbium also has the 631 nm transition with a linewidth of 28 kHz and a Doppler temperature of 680 nK and the 841 nm transition with a linewidth of 8 kHz and a Doppler temperature of 190 nK.

The capture velocity of a MOT is the maximum velocity that an atom can have to still be captured in the MOT rather than simply being slowed a little but still escaping the cloud. We can use basic kinematics to estimate this capture velocity, assuming a constant acceleration, and where D is the distance over which this acceleration is felt by the atoms :

$$v_c^2 = 2 * \text{acceleration} * D$$

If we assume the atoms feel the maximum possible force F over the entire diameter D of the MOT beams, then the acceleration is $F/m = \hbar k\Gamma/2m$. This gives an estimate of the capture velocity v_c for any given transition as [38]:

$$v_c = \sqrt{\frac{2\hbar k\Gamma}{2m} D}$$

For the 583 nm transition, with a MOT beam diameter of 25mm, the capture velocity is about $v_{c,583} = 5$ m/s. For the 631 nm transition, $v_{c,631} = 1.5$ m/s, and for the 841 nm line, $v_{c,841} = 0.75$ m/s. The 583 nm line is still wide enough that it is possible to load a MOT directly from a Zeeman slower on this transition. The 631 and 841 nm transitions however are a bit more difficult, it would be difficult to slow atoms down to these velocities with a Zeeman slower. Because of this, we operate our experiment by first loading atoms from the Zeeman slower into a 583 nm MOT,

compressing and cooling the atom cloud down to (or nearly to) the Doppler temperature, and then handing off the slowed atoms to a MOT on the narrower 841 nm transition. Once cooled in the first-stage 583 nm MOT, the atoms are slowed down to well-below the 0.75 m/s capture velocity of the second-stage 841 nm MOT. In fact, the Doppler velocity of the 583 nm transition is 15 mm/s so even without reaching the Doppler temperature, the atoms should be able to be transferred into the 841 nm MOT.

The 583 nm light is generated from a Toptica DL-Pro at 1166 nm seeding an MPB Raman Amplifier, see [27] for more detail on the laser system, the optics setup and the frequency stabilization. We operate the 583 nm MOT with very large beams, approximately 5 cm in size, nearly filling the optical access given to us by the viewports of the vacuum chamber. We use two retro-reflected horizontal beams, and one retro-reflected vertical beam. We incorporated frequency scanning of the light, as in [33], to broaden the frequency spectrum of the light to about 4 MHz. We found this helped increase the loading rates.

Due to a combination of the low radiation pressure on the atoms from the MOT light and the large mass of the erbium atoms, the equilibrium position sags to a position with higher magnetic field. This causes the atoms to preferentially scatter light from the upwards-propagating $\sigma-$ vertical beam rather than from the retro-reflected $\sigma+$ downwards-propagating MOT beam. This causes the atoms to be polarized, since the light is pumping the atoms into the lowest Zeeman sub-level $m_J = -6$. This spontaneous polarization of the atom cloud is a very convenient feature of the lanthanide narrow-line MOTs, previously seen in the dysprosium 626 nm MOTs [39] and [34], as well as the erbium 583 nm MOTs [21].

We verified experimentally that the atom cloud is spin-pure via Stern-Gerlach measurements. For this, we used our in-vacuum, gradient coil, which is described in more detail in Chapter 5.5.

The 583 nm MOT is loaded in the main chamber, as seen in Figure 2.2. Typical parameters for loading the 583 nm MOT are a gradient of 2.3 G/cm, an optical intensity of 20 Isat, and a detun-

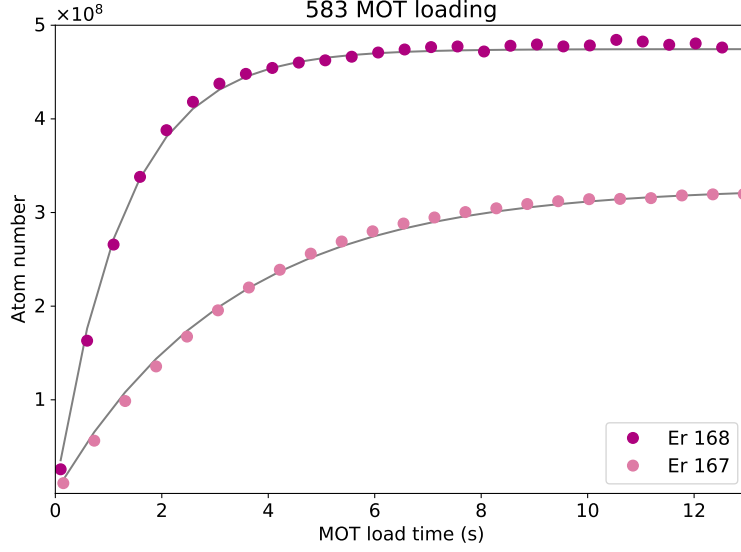


Figure 3.1: MOT loading of two isotopes, bosonic ^{168}Er and fermionic ^{167}Er . Fitting equation 3.1 to the data, we extract the load rates of these two isotopes: $R_{168} = 36(8) \times 10^6$ atoms/sec and $R_{167} = 10(1) \times 10^6$ atoms/sec.

ing of -6 MHz, or about 30Γ , as shown in Figure 3.3. Due to the large size of the MOT beams and the careful design of the 2D molasses section, the 583 nm MOT loads very quickly for the more abundant isotopes. For example, during a typical run of our experiment, we load 50×10^6 atoms of ^{168}Er in only 200 ms. See Figure 3.1 for 583 nm MOT loading curves of ^{168}Er (bosonic) and ^{167}Er (fermionic). Table 3.1 shows the fitted load rates and other extracted parameters for the 583 nm MOT of ^{168}Er , compared to the parameters for the 841 nm MOT of ^{168}Er .

To extract the loading rate of the atoms in the MOT, we use the standard loading equation:

$$N(t) = N_{ss}(1 - e^{-\gamma t}) \quad (3.1)$$

where the steady state atom number in the MOT is given by $N_{ss} = R/\gamma$ and R is the capture rate and γ is the decay rate, as in [33].

Once the atoms are loaded into the 583 nm MOT, we start a compression sequence (cMOT)

which allows us to increase the density and reduce the temperature of the atom cloud. To understand the mechanism behind the compression and cooling of the atoms in the MOT, we must look at some theory of narrow-line cooling. Following [28] and [40], we can start by looking at the radiative force equation in 1 dimension, along z , which is:

$$F_{\text{MOT}} = \frac{\hbar k \Gamma}{2} \left(\frac{s}{1 + s' + 4(\Delta - kv_z - \mu(\partial_z B_z)z)^2/\Gamma^2} \right) - mg, \quad (3.2)$$

where $k = 2\pi/\lambda$ is the wavevector of the MOT light, Γ is the decay rate of the atomic transition, $s = I/I_{\text{sat}}$ is the saturation parameter, where I is the intensity of the MOT beams. Here, s' is a parameter that accounts for saturation effects due to light from the MOT beams in the other directions. The force depends on the saturation parameter, the detuning of the light Δ and the magnetic field gradient $(\partial_z B_z)$.

The equilibrium temperature of the atoms in the MOT is determined by the ratio of the heating (caused by diffusion) and the cooling (caused by damping) rates. In the end, this gives an equilibrium temperature T_{equil} which is independent of the detuning of the MOT light, but depends on the intensity of the light:

$$T_{\text{equil}} = \frac{\hbar \Gamma \sqrt{s}}{2k_B} \frac{R}{2\sqrt{R - s'/s - 1/s}}, \quad (3.3)$$

with $R = \hbar k \Gamma / 2mg$. From this dependence of the equilibrium temperature on the saturation parameter, we can see that lowering the MOT beam intensity lowers this T_{equil} and cools the atom cloud. For our typical parameters, this gives a $T_{\text{equil}} = 160 \mu\text{K}$ for the equilibrium temperature during the MOT loading with an intensity of $s = 20I_{\text{sat}}$, and $T_{\text{equil}} = 16 \mu\text{K}$ for the equilibrium temperature at the end of a typical compression of our 583 nm MOT down to an intensity of $s = 0.2I_{\text{sat}}$. This is a typical compression sequence we use before loading into the second-stage 841 nm MOT, and does not reflect the fundamental temperature limits of a MOT operating on this transition. We have in fact measured temperatures down to about 5 μK in this MOT.

Er ₁₆₈ 583 and 841 MOT measurements		
	583 nm MOT	841 nm MOT
Load rate R ($\times 10^6$)	36(8)	31(7)
Decay rate γ (s^{-1})	0.7(7)	0.7(5)
Steady-state atom number N_{ss} (10^6)	47(4)	42(0)
Lifetime (s)	0.4(70)	0.23(3)

Table 3.1: Lifetime and loading parameters for the 583 and the 841 MOT of ¹⁶⁸Er. Steady-state atom number N_{ss} and decay rate γ are extracted from Equation 3.1. Load rate $R = \gamma N_{ss}$. The lifetimes are given by the 1/e time of the exponential decays fitted to the hold time data.

The second ingredient in the compression of the MOT is reducing the size of the atom cloud. In 3D, the radiative force in Equation 3.2 forms ellipsoidal shells of constant force, centered around the magnetic field zero. The atoms sit on the shell of $F_{MOT} = 0$. However, as mentioned above, they sag down due to gravity and the narrowness of the line and therefore sit at the bottom of this ellipsoid. By reducing the detuning, the shell of zero force shrinks and the atoms are raised towards the magnetic field zero. The shell shrinking also causes the atom cloud to shrink in size, and hence the atom density to increase.

The lifetime of the atoms in the 583 nm MOT is shown in Figure 3.2. Due to the very high degree of polarization of the atom cloud, we don't expect any spin-changing 2-body collisions in the cloud. When fitting the data to a double exponential we extract two distinct timescales in the decay. A rapid decay with $1/\gamma_{fast} = 4(70)$ ms and a slower decay with $1/\gamma_{slow} = 2.(6)$ s. The rapid decay of the atoms in the 583 nm MOT measured here is consistent with the lifetime that was measured in the erbium MOT in [41].

3.2 841 MOT

Once the atoms have been trapped and cooled in our first 583 nm MOT, we transfer them into a MOT operating on the narrower, 8 kHz wide transition at 841 nm.

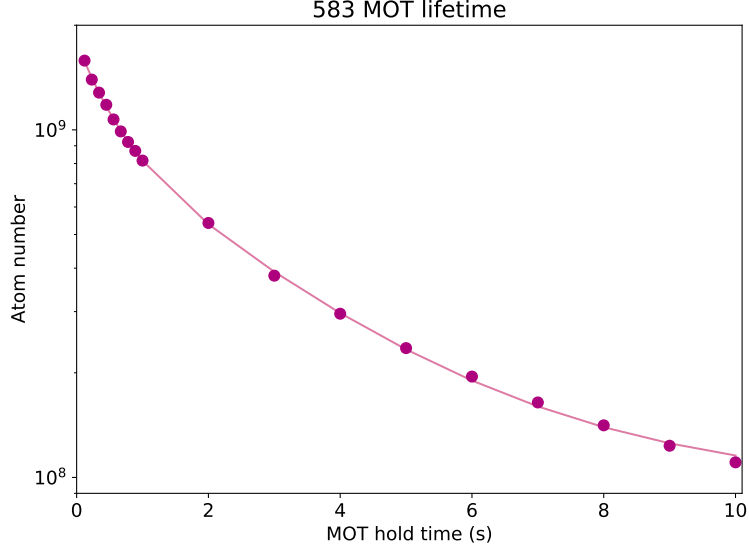


Figure 3.2: Lifetime of the atoms in the 583 MOT, at a detuning of $8 \Gamma_{583}$, or 1.5 MHz and a beam intensity of $0.2 I_{\text{sat}}$. A double exponential is fit to the data from which we extract a rapid decay with a timescale of 4(70) ms and a longer decay with a timescale of 2.(6) s.

We can estimate the velocity of the atoms in the 583 nm cMOT (after the cooling due to the compression step) by calculating the thermal velocity of the atoms at 15 μK :

$$v_{\text{thermal}} = \sqrt{\frac{k_B T}{m}} = 0.02 \text{ m/s}$$

This velocity is well below the capture velocity for the 841 nm MOT which we estimated above to be about $v_{c,841} = 0.75 \text{ m/s}$. This is a first sign that the transfer between the two MOTs should be successful.

The hand-off can be optimized by matching the potentials of the two traps, and simply turning the 583 nm light off at the same time as turning on the 841 nm light. The atoms should then go from being confined by the trap formed with the 583 nm light to that formed by the 841 nm light. We can estimate the potential of the compressed 583 nm MOT so as to find a range of parameters for the 841 nm MOT light intensity, detuning, and magnetic field gradient which give us good over-

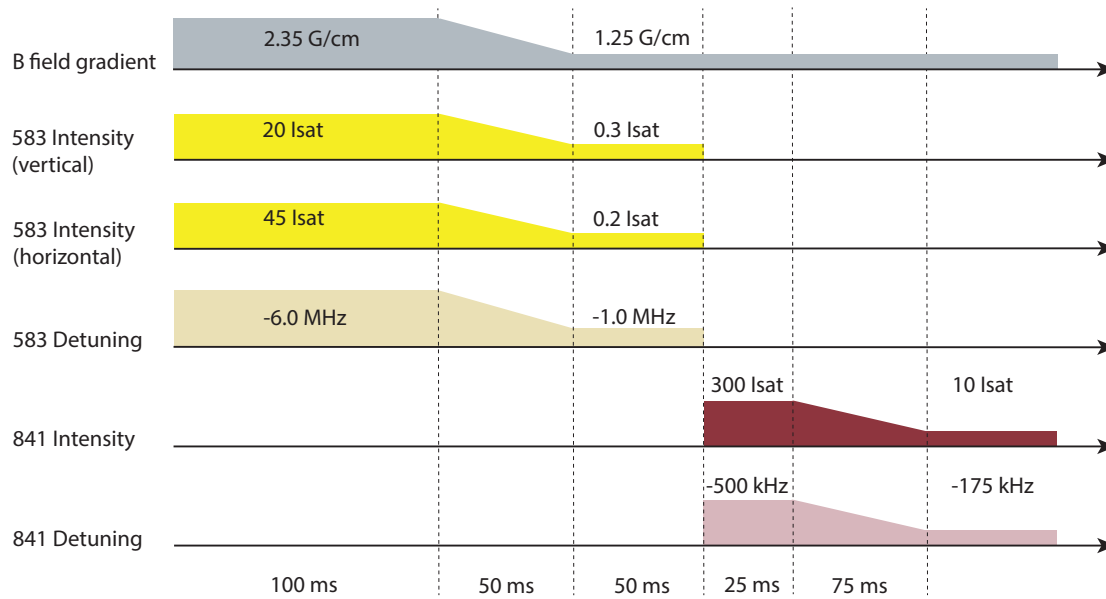


Figure 3.3: Experimental sequence for the loading and compression of the 583 nm MOT, the hand-off into the 841 nm MOT and the compression of the 841 nm MOT. The parameters shown are for a typical run of the experiment generating a cold cloud at about $0.5\mu\text{K}$, of ^{168}Er .

lap of the two potentials. This will ensure a maximum number of atoms that are “dropped” from the 583 nm MOT will be “caught” by the 841 nm MOT.

The potential can be estimated by simply taking the gradient of the force in Equation 3.2 and using the experimental parameters for the operation of the 583 nm MOT that are shown in Figure 3.3. To give a sense of scale of the compression of the first stage MOT, the resulting potential for the 583 nm MOT during loading and after compression are shown in Figure 3.4. The depth of the 583 nm MOT during loading is more than 500 mK, and after compression, is on the order of 10 mK.

The hand-off potentials of the two MOTs are shown in Figure 3.5, as well as the potential of the 841 nm MOT at the end of its compression. The depth at the moment of hand-off is around 15 mK, with the 841 nm trap being slightly deeper than the initial 583 nm trap. The estimated potentials are also very well matched spatially for an efficient hand-off.

To facilitate the hand-off between the two traps, and to maximize the mode-matching of the two

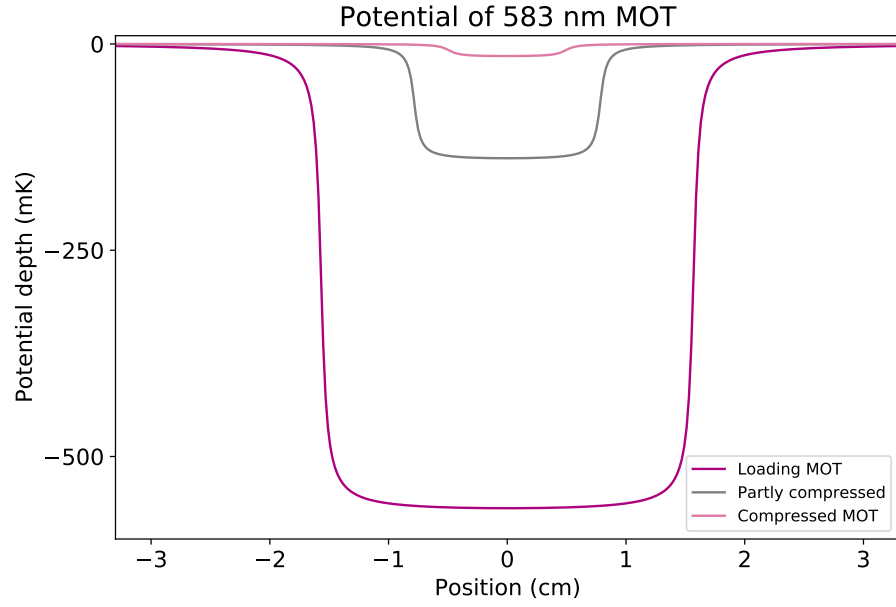


Figure 3.4: Theoretical estimate of the potential of the 583 nm MOT during loading and after the compression are shown here. The MOT potential during loading is quite deep (more than 500 mK) and the cloud is large. The partly compressed MOT curve is estimated for a detuning of $\Delta = 15\Gamma_{583}$, intensity of $s = 5I_{\text{sat}}$. The loading MOT and compressed MOT curves are estimated for the loading and compression of a typical sequence, for the parameters shown in Figure 3.3.

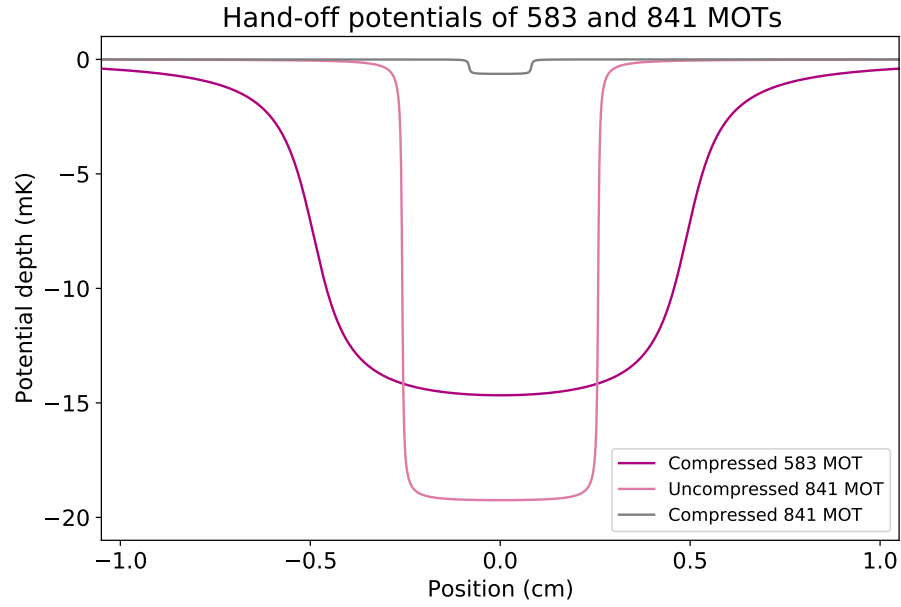


Figure 3.5: Plot showing theoretical estimates of the potentials of the 583 nm MOT and the 841 nm MOT at the time of the hand-off. The atoms are initially trapped in the compressed 583 nm MOT. Simultaneously the 583 nm light is turned off and the 841 nm light is turned on, transferring the atoms to the uncompressed 841 nm MOT. The 841 nm MOT is then compressed. The final 841 nm potential (shown in grey) is extremely shallow and narrow, allowing us to achieve cold temperatures and high phase-space densities. Parameters for these curves are the same as experimental parameters shown in Figure 3.3.

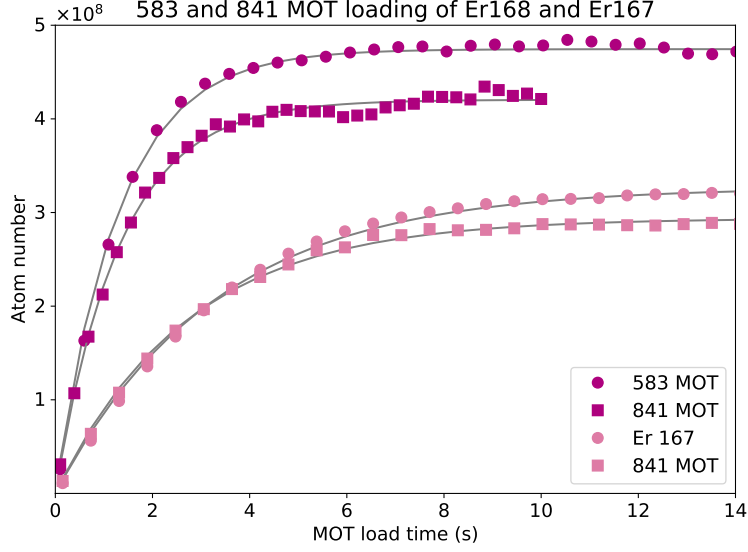


Figure 3.6: Transfer of atoms from 583 nm MOT to 841 nm MOT, with increasing 583 nm MOT loading time on the x -axis. For short 583 nm MOT load times, which means for smaller initial atom numbers, the transfer between the 583 nm MOT and the 841 nm MOT is essentially 100% efficient. For larger initial atom numbers, the efficiency drops as the 841 nm MOT saturates before the 583 nm MOT does. This lower efficiency starts to be apparent at around 1s for the ^{168}Er isotope and around 4s for the fermionic ^{167}Er isotope.

colors of MOT beams with easiest alignment, the two colors are combined with a dichroic mirror right after their respective fibers. In this way, the 583 nm and the 841 nm light share a beam path through the lenses expanding and collimating the beams, the polarization optics, and the beam-steering mirrors. The layout of the MOT optics is shown in detail in [27].

The only custom optic in this set-up are the quarter waveplates used to create the circularly-polarized light necessary for operating the MOT. The waveplates, from Knight Optical, were custom designed to provide $\lambda/4$ of retardation at both 583 nm and 841 nm. They are coated with an anti-reflection coating, with a reflection at both the wavelengths of interest of $R < 0.5\%$. The waveplates are 2" in diameter, and about 0.75 mm in thickness.

All the above comes together to give us a reliable, very efficient transfer of the atoms. For low atom numbers in the compressed 583 nm MOT, the transfer is actually close to 100% efficient,

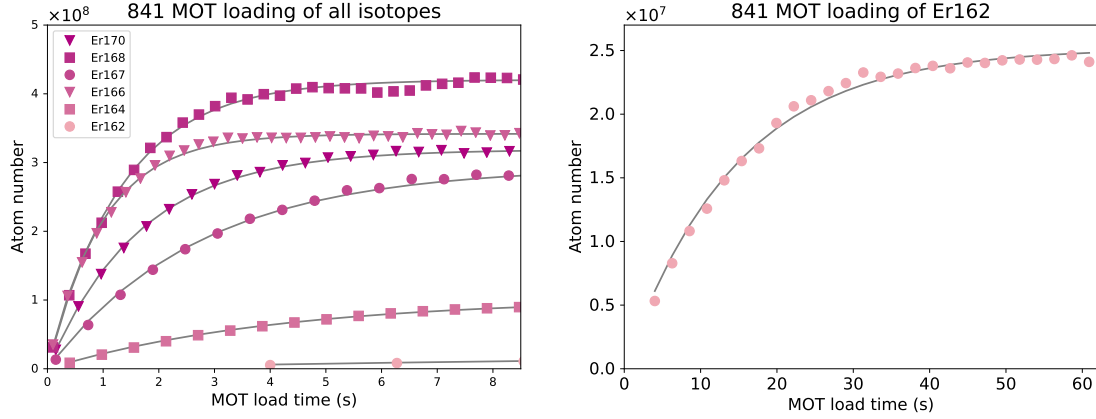


Figure 3.7: Left: Loading of the 841 nm MOT from the 583 nm MOT for all 6 erbium isotopes. Right: loading of the 841 nm MOT from the 583 nm MOT for ^{162}Er , plotted again separately due to load rate being significantly slower than other isotopes. Loading rates are extracted by fitting the data to equation 3.1. See Table 3.3 for resulting loading rates.

with no measurable drop in the atom number in the 841 nm MOT. The atom number in the compressed 583 MOT right before transfer and the atom number captured in the 841 MOT are plotted together in Figure 3.6 for a bosonic (Er168) isotope and the fermionic (Er167) isotope. For longer load times of the 583 MOT (or, higher initial atom number before the hand-off) the transfer becomes less efficient due to the 841 MOT saturating at a lower atom number than the 583 MOT. This tells us that later in the sequence our atom number can only be increased by increasing the 583 MOT loading time up to the saturation of this 841 MOT.

The transfer into the 841 MOT for all stable isotopes of erbium is shown in Figure 3.1. For most isotopes, the 841 nm MOT saturates around 300×10^6 atoms within several seconds. The most abundant isotope, Er168 , we can obtain 400×10^6 within about 3 seconds, and reach 100×10^6 in less than half a second. As for the loading of the 583 nm MOT, we fit the atom number data to Equation 3.1 to extract the load rates and the saturation values for each isotope. The results are in Table 3.3.

It is interesting to note that the loading rates of each of the isotopes are roughly consistent with their relative abundance. We can calculate for each isotope the ratio of its loading rate relative to the

Loading rates compared to isotope abundance		
Isotopes	R/R_{166}	NA/NA_{166}
162	0.003	0.003
164	0.07	0.05
166	1	1
168	0.9	0.8
170	0.5	0.4

Table 3.2: The MOT loading rate R and natural abundance NA of each bosonic isotope, compared to the values for ^{166}Er . The similarity between the numbers in each column indicates the lower natural abundance of certain isotopes could be the sole reason for the slower loading rates. See text for comment about the fermions.

loading rate of ^{166}Er . We can also calculate the ratio of its natural abundance relative to the abundance of ^{166}Er . By comparing these two values, we find that the natural abundance is sufficient to explain the difference in loading rates. The relative abundance of the isotopes is found in Table 2.1.

For example, with ^{164}Er , the ratio of the loading rates R is: $R_{164}/R_{166} = 0.07$ And the ratio of the natural abundances NA is: $NA_{164}/NA_{166} = 0.05$. This is summarized for all the bosonic isotopes in Table 3.2. The spectroscopy measurements on the 401 nm transition, we observed that for the fermionic isotope, ^{167}Er , the atoms in the atom beam populate a spread of several of the sublevels (19/2 dominates, then 17/2, then 15/2 etc). For this reason, so the MOT loading rate is subsequently lower than one would have naively expected from the natural abundance of the isotope.

We also investigate the lifetime of the atoms in the 841 nm MOT. The atoms are held in the uncompressed MOT and the atom number is measured for increasing hold times, for all stable isotopes, as shown in Figure 3.8. All isotopes experience a rapid initial decay with a timescale of 0.1-0.5 seconds. This decay is relatively quick, however for our experiment this MOT serves only to transfer into an optical-dipole trap. This transfer is done in the order of 100 ms, so the atoms don't spend enough time in the 841 nm MOT for this decay to be a problem.

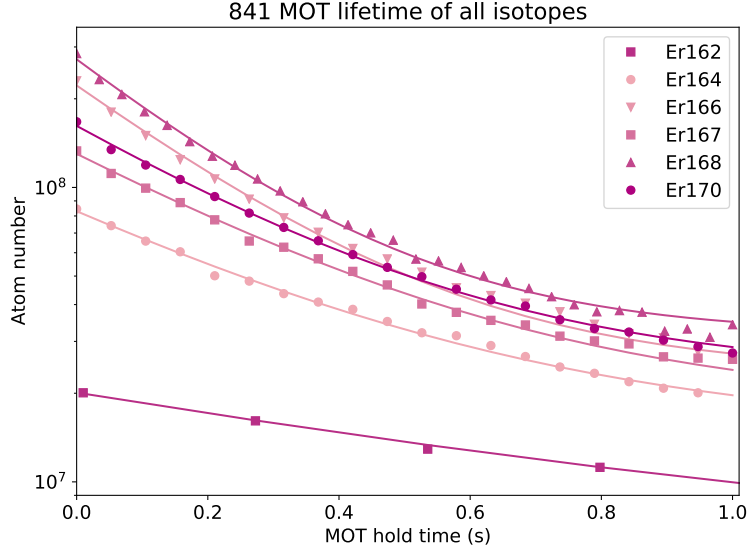


Figure 3.8: Lifetime of the atoms in the 841 nm MOT. Data is fit to a single exponential, and decay rates are in Table 3.3. All isotopes experience a fast decay with a timescale on the order of 0.1-0.5 seconds.

The decay rates are extracted by fitting the lifetime data in the 841 to an exponential decay. The resulting lifetimes for all stable isotopes are found in Table 3.3.

In the same way as for the 583 nm MOT, we verified the spin purity of the atom cloud in the 841 nm MOT via Stern-Gerlach measurements. We rapidly pulsed on a large gradient and verified that the whole cloud is shifted, with no increase in size that would indicate a population in higher spin states. This tells us the atoms are still all in the $m_J = -6$ state after the transfer.

The compression of the 841 nm MOT is similar to that of the 583 nm MOT. We reduce the intensities and the detunings to decrease the temperature and increase the density of the cloud. One difference is that we found no advantage to further decreasing the magnetic field gradient during the 841 compression sequences, whereas in the 583 compression we reduce the gradient by about a factor of two, from 2.35 G/cm to 1.25 G/cm. In a typical compression, the detuning is decreased from $60\Gamma_{841}$ to $20\Gamma_{841}$ (or, from about 500 kHz to 175 kHz). The intensity is decreased from 300 I_{sat} to about 20 I_{sat} , leading to a temperature of about 700 nK. See Figure 3.3 for a schematic of the

841 MOT measurements for all isotopes						
	162	164	166	167	168	170
Load rate R (10 ⁶ atoms/s)	1.(0)	24.(0)	34(4)	10(8)	31(7)	19(2)
Decay rate γ (s ⁻¹)	0.06(9)	0.24(3)	1.0(1)	0.3(6)	0.7(5)	0.6(0)
Steady-state atom number N _{ss} (10 ⁶)	25.(1)	10(2)	34(1)	29(3)	42(0)	31(9)
Lifetime (ms)		3(70)	25(0)	3(30)	23(3)	3(10)

Table 3.3: Measured lifetimes, load rates R, decay rates γ and steady-state atom numbers $N_{ss} = \gamma \times R$ for the 841 nm MOT of each stable isotopes of erbium. The lifetime data for the least abundant isotope ¹⁶²Er is incomplete, for no reason other than not enough data was collected to get a reliable number on this isotope's lifetime.

full sequence of both MOTs.

The narrow-line cooling theory is the same as described above for the 583 nm MOT, giving us the same dependence of the MOT's equilibrium temperature on the intensity of the beams. We verify this dependence experimentally by compressing the 841 nm MOT to varying MOT beam intensities, and measuring the temperature. For each beam intensity, we extract the temperature of the cloud by varying the length of the time-of-flight. The longer the time-of-flight, the more the cloud expands, and then fitting to the size of the cloud. This gives us a temperature measurement, which we then repeat for each beam intensity. The result is shown in Figure 3.9, plotted against the theory given by Equation 3.3. The theory is with $s'=s$, with no free parameters. Our range of temperatures measured in this plot are limit by the available power in our system on the upper end, and by the dynamic range of our intensity-stabilization PID for the lower end. We cannot reliably reach lower beam intensities than about 1 Isat with our current system, so we are unable to cool the MOT further than about 300 nK.

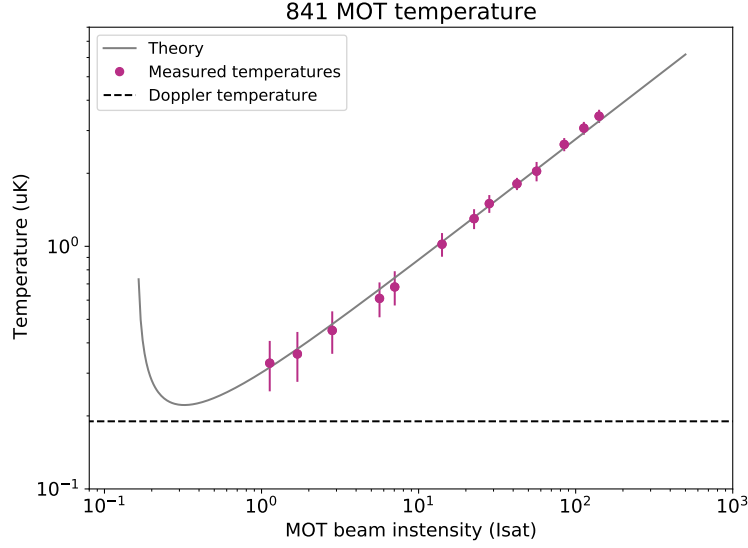


Figure 3.9: Theoretical equilibrium temperature for the 841 nm MOT, with measured temperatures. This shows the dependence of the equilibrium temperature on the beam intensity. The lower temperature limit shown is the Doppler temperature of the transition. The theory is simply Equation 3.3, with no free parameters.

3.3 SPECTROSCOPY

Our use of the narrow, 841 nm atomic transition of erbium gives us the opportunity to measure the isotope shift of this transition between the 6 different stable isotopes. Previously, much work has been done to identify and measure the transition frequencies of the 401 nm, 583 nm and 841 nm transitions in each isotope, as in [42], [43], [44] and [29]. Here I will only discuss our measurements of the 841 nm transition frequencies in all 6 isotopes. See [27] for our measurements of the 583 nm and 401 nm frequencies, using similar techniques.

We use a “push beam” method for our spectroscopy measurements. The basic principle is that we drop a cloud of cold atoms from the 841 nm MOT and briefly illuminate the cloud with a pulse of 841 nm light (we call this our “push beam”) from one direction. The cloud of atoms will then feel a horizontal momentum kick from this pulse of light, which is dependent on the detuning of the light relative to the atomic transition. This momentum kick will displace the atom cloud along

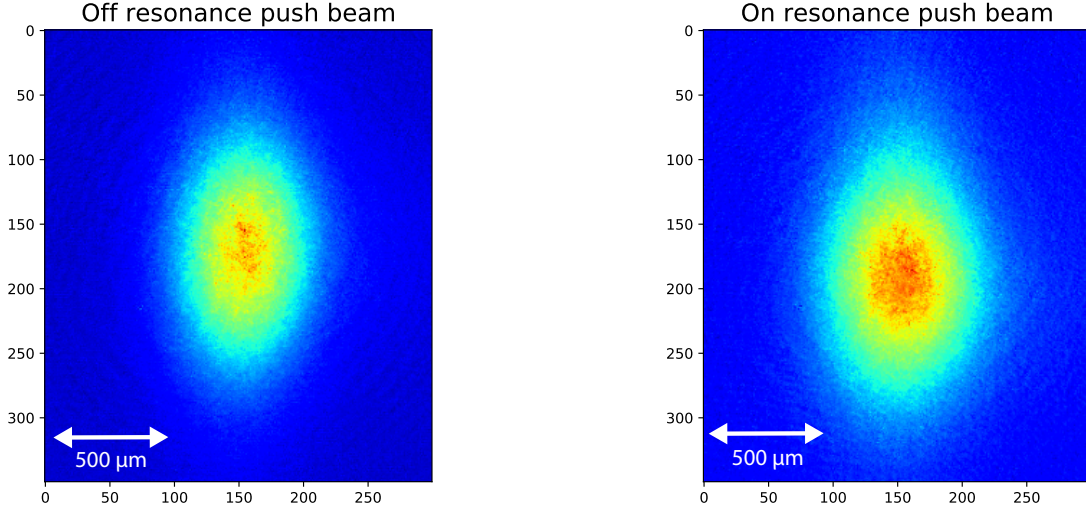


Figure 3.10: Absorption images of the 583 nm MOT after 34 ms of time-of-flight, 4 ms of which are during the push beam pulse of 841 nm light. On the left is an image of the cloud after an off-resonance push beam at a detuning of about 100 kHz, or about $12\Gamma_{841}$ from the resonance. On the right is an image of the cloud after an on-resonance push beam, within $1\Gamma_{841}$ from the transition. With our push beam parameters, the cloud center is pushed by a maximum of about 0.2 mm relative to the off-resonance position.

the direction of the “push beam” as the cloud continues to fall during the time-of-flight before the absorption image is taken. The amount of displacement of the cloud can be extracted by fitting to the absorption image, and depends on the light’s detuning and intensity and the duration of the pulse. If we keep the second two variables constant and repeat measurements while varying the detuning of the light, we can then extract the cloud’s displacement as a function of detuning.

The force felt by the atoms from this “push beam” is a Lorentzian in the detuning of the light from the atomic transition, with the maximum force occurring when the light is on resonance. We can reduce the intensity and the length of the pulse so that the atoms only feel a small force and the cloud is not blown away completely or heated up dramatically. Figure 3.10 shows a side by side of an absorption image of the cloud after an off resonance push beam and an absorption image after an on resonance push beam.

We fit the measurements of the cloud center to a Lorentzian, shown in Figure 3.11. This allows

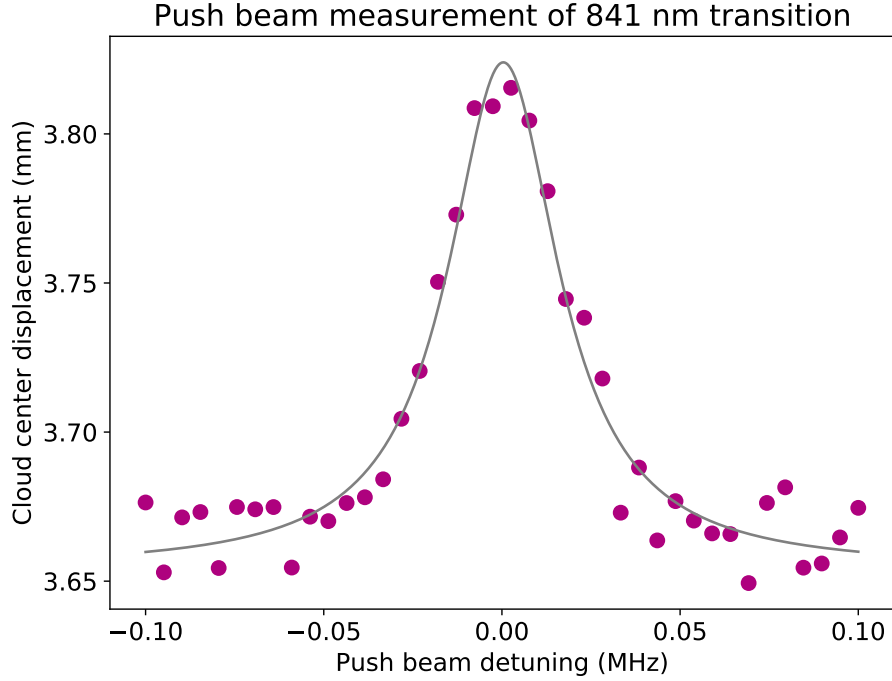


Figure 3.11: Center position of the atom cloud for varying push beam detunings. The atoms are dropped from the 583 nm MOT for 34 ms and are pushed with a 4 ms long pulse of 841 nm light during the time-of-flight. The cloud center is extracted from the absorption image and plotted here. The grey line is a Lorentzian fitted to the cloud center data. The Lorentzian peak is centered within $1 \Gamma_{841}$ of 0, indicating our frequencies are properly calibrated.

us to extract the peak of the cloud center’s displacement, and find the push beam frequency corresponding to the atomic transition.

We can find the transition frequency for each isotope, and calibrate the AOM frequencies accordingly. We can then calculate the shift of the transition of each isotope relative to one isotope, here Er_{166} , by knowing the free spectral range (FSR) of the cavity that we are locking the 841 nm laser to. Our cavity has an FSR of $1498.08(5)$ MHz. The resulting measurements of the isotope shifts are found in Table 3.4.

The push beam measurements are performed on atoms from an 841 nm MOT, which allows us to reach colder temperatures than in the 583 nm MOT. We load 10 million atoms, which are then compressed to $6 \Gamma_{841}$ to create a dense cloud, and are cooled to about $2 \mu\text{K}$.

Spectroscopy measurements of 841 nm transition	
Isotope	Frequency (MHz)
162	- 3,254.4
164	- 1,424.1
166	0
167	238.2
168	1,434.9
170	2,913.7

Table 3.4: Measured frequencies of the 841 nm transition for all stable isotopes of erbium. Measurements are given relative to the transition frequency of the ^{166}Er isotope. See [30] for more details.

To minimize the sources of error and noise as much as possible, we have to be a bit careful about several things. One is the intensity and length of the push beam (these are equivalent as they both contribute to the total number of photons absorbed by the cloud). If the intensity is too high, the profile shown in Figure 3.11 becomes power-broadened and no longer fits well to a Lorentzian, which makes it harder to accurately measure the center of the resonance. If the intensity is too low, the magnitude of the cloud's horizontal displacement becomes too small to measure reliably. We settled on a compromise in intensity of the push beam resulting in a shift of the cloud of about 0.2 mm. This is with a push beam intensity of a few I_{sat} , lasting 4 ms.

The atoms undergo 34 ms of free evolution during the time-of-flight. It is important that the magnetic field on the atoms be carefully zeroed during this free fall. Any stray fields will allow the push beam pulse to drive transitions to other magnetic sublevels, which will alter the cloud's frequency response. Driving multiple transitions causes the feature shown in Figure 3.11 to split into multiple peaks. If the magnetic field is large enough, then the individual peaks are resolvable, but if the magnetic field is smaller, then this manifests as a single, broadened peak. In order to surmount this, we calibrated our magnetic field along each axis, using the shape and width of the Lorentzian feature. More details on the analysis can be found in [30].

4

Fast production of quantum gases

The ability to create quantum degenerate gases has brought us platforms for studying few- and many-body quantum phenomena. However, the complexity of some of the questions that can be answered by these systems is limited by the number of measurements that can be reasonably performed given the current standard of measurement rates. The limiting factor is typically the sample preparation time of the degenerate gas. In many quantum gas microscopes, for example, the cycle time of the experiment is on the order of 10-100 s, of which almost the entire duration is spent on

slowing and trapping the atoms and cooling them to degeneracy.

Reducing the sample preparation time to 0.1-1 s would allow for vastly more data to be taken and would expand the scope of problems that are possible to study on these cold-atom platforms. Recent work has made significant progress in speeding up the production of degenerate quantum gases. All-optical methods have shown much promise, as in [45], [46], [47], [48]. Additionally, work optimizing evaporation sequences, using dynamic traps, enabling efficient evaporative cooling, has also proven effective [49].

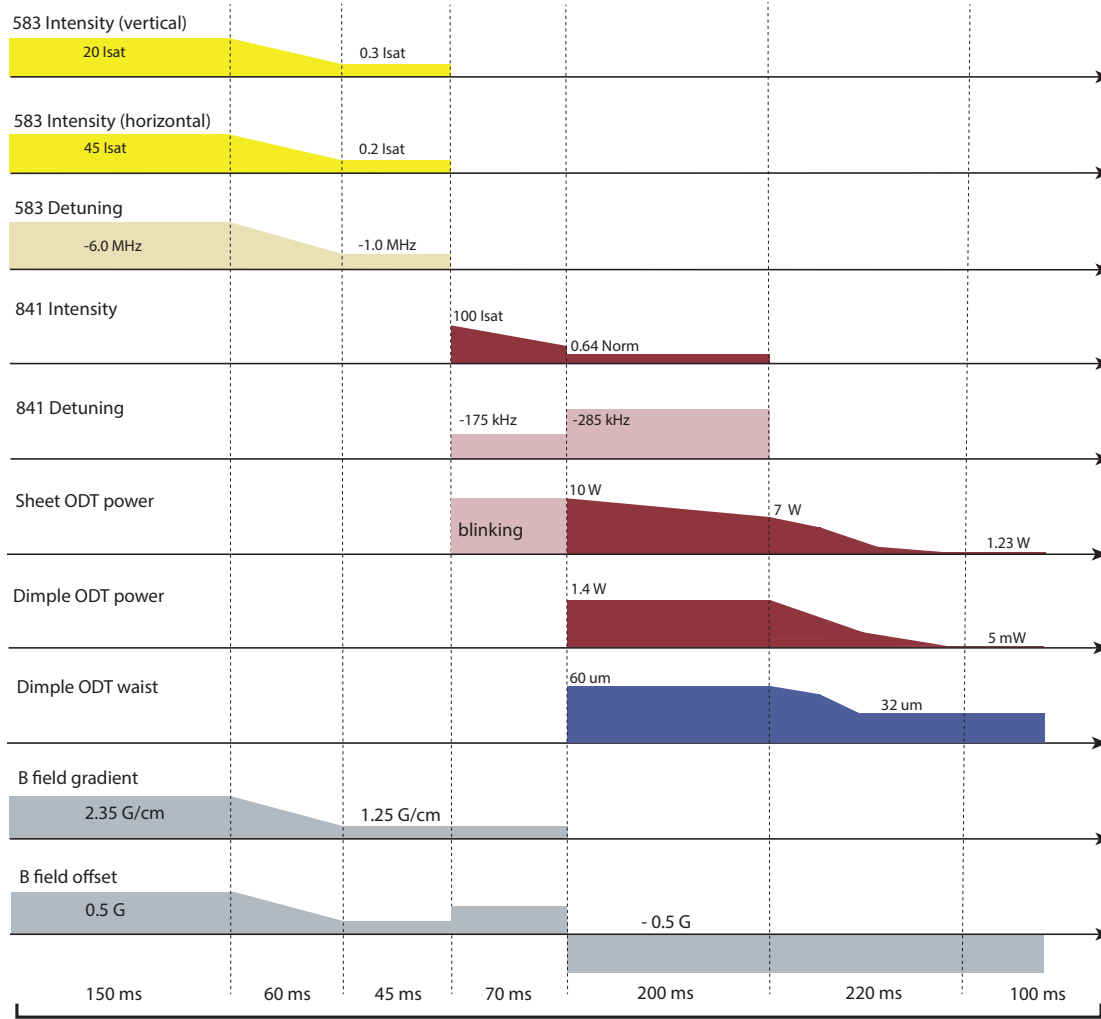
Here, we bring the BEC preparation time down to 850 ms for a nearly pure BEC of 80,000 atoms. For the fermions, we create a DFG in about 4 s. This is a significant improvement on the sample preparation time of other quantum gas microscopes, bringing us to repetition rates comparable to that of Rydberg tweezer arrays [50], [51], [52]. To achieve this, we start off with a MOT with an extremely high load rate, almost 400×10^6 atoms/sec as discussed in Chapter 3. A very short MOT load time, combined with a tunable optical dipole trap geometry and the advantages brought to us by the narrow-lines of erbium, has allowed us to reach these very short preparation times for large erbium BECs and DFGs.

The results of this chapter are published in [53].

4.1 OVERVIEW OF CYCLE TO DEGENERACY

Here we start with a brief overview of the experimental sequence used to create our degenerate gases of erbium. Subsequent sections go into more detail of each of the more interesting steps of the sequences.

In creating Bose-Einstein condensates and degenerate Fermi gases of erbium, we start with the atoms in a compressed 583 nm MOT, as described in Section 3.1. We then transfer the atoms into an 841 nm MOT, which we compress to a phase space density estimated at 0.05. At the same time



Pure BEC in ~850 ms

Figure 4.1: Experimental sequence for the creation of a pure BEC of Er_{168} in 850 ms. The parameters shown are for a typical run of the experiment generating a BEC of 10^5 atoms.

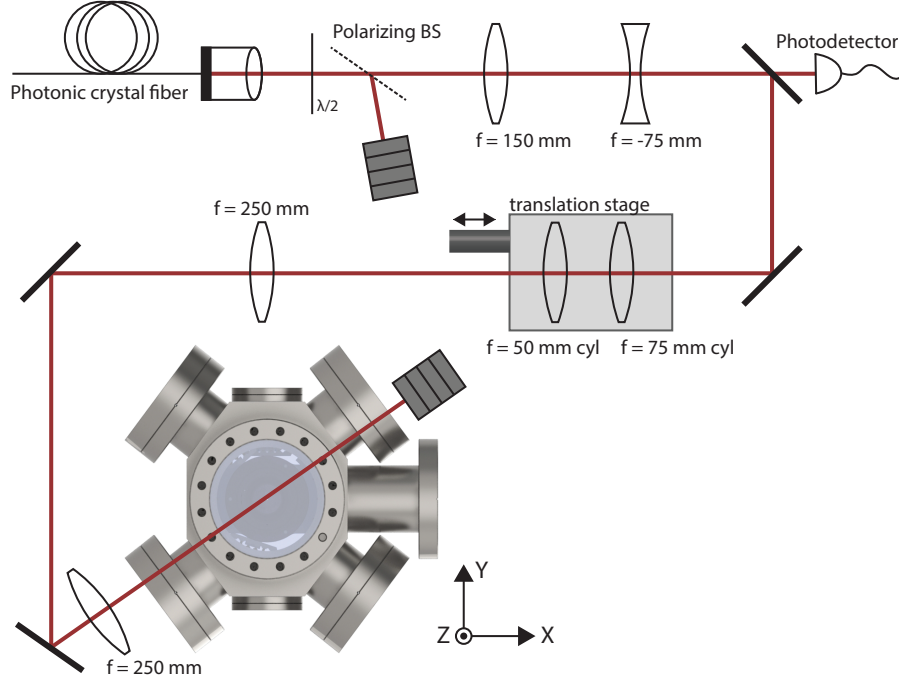


Figure 4.2: Diagram showing the optical layout for the sheet ODT beam path, after the photonic crystal fiber.

as the atoms are cooled in the 841 nm MOT, the first stage of the optical dipole trap (ODT) is turned on, allowing atoms to simultaneously start loading into the first, horizontal ODT, the “sheet ODT.” The next stage is to turn on a small, vertical ODT, and apply a cooling molasses pulse of 841 nm light. A short evaporation sequence follows, in which the trap frequencies and trap depths are lowered until the sample reaches degeneracy. A brief hold time brings the purity of the BEC up. The experimental cycle described here is summarized in Figure 4.1.

4.2 OPTICAL DIPOLE TRAPS

4.2.1 HORIZONTAL SHEET ODT

The “sheet ODT” is a large horizontal beam that we use to first trap the atoms after the 841 nm MOT. The ODT light is from a 45 Watt, 1064 nm fiber laser from Azur Light Systems. We use a

2 meter photonic crystal fiber from NKT Photonics to deliver the light to the main chamber. The reason for using a fiber is to isolate the two optical setup. In this way, miss-alignments (for example in the AOMs which are needed for power control) do not affect the alignment of the ODT beam on the atoms in the chamber.

The sheet ODT beam is highly elliptical so as to maximize the overlap of the trap with the compressed 841 nm MOT, to ensure the highest transfer possible. The beam waists on the atom are $550\mu\text{m} \times 19\mu\text{m}$. The large waist in the horizontal axis matches the elongated shape of the atom cloud, and the tight waist in the vertical axis provides tight confinement to better hold the atoms against gravity. The optical layout for the sheet ODT beam after the fiber on the main chamber side is shown in Figure 4.2. The initially circular beam is expanded in the vertical axis by two cylindrical lenses, as shown in the optics layout. This axis is then focused down to a tighter focus, giving us the high aspect ratio in the beam.

The sheet ODT is first turned on to a power of 11 Watts, while the 841 nm MOT is being compressed. However the 1064 nm light induces a differential AC Stark shift on the excited 841 nm state, causing the atoms that are in the presence of this high-power 1064 nm ODT beam to not feel the cooling effects of the 841 nm MOT beams. See [54] and [55] for an overview of atomic polarizability.

We estimate that the excited state polarizability is approximately 60% that of the ground state. To solve this challenge, we implemented a system that strobes the ODT light on and off during the time that both the 841 nm MOT and the sheet ODT are on. We call this the “blinking” ODT.

This blinking allows the atoms to alternate between periods of time during which they are trapped by the ODT potential, and periods of time during which they are cooled in the MOT. In order for both of these regimes to be at least partially in effect, there are a few timescales to keep in mind. First, the atoms need to see the unshifted 841 nm light (with no 1064 nm light on) for long enough to scatter 841 nm photons in order to be properly cooled by the MOT. This means the blinking

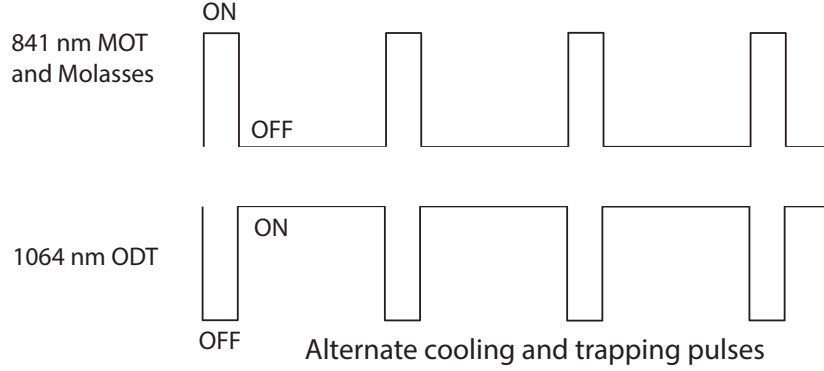


Figure 4.3: Diagram illustrating the blinking sequence of the 1064 nm sheet ODT and the 841 nm MOT during the loading of the atoms from the MOT into the ODT. The strobing happens at a frequency of 5 kHz, with an 80% duty cycle with the ODT light on for 80% of the time.

must be slower than the scattering rate of the 841 nm excited state. Secondly, the atoms must still on average feel the potential from the ODT in order to remain trapped. This means that the strobing on and off of the ODT light must be faster than the vertical trap frequency. The atoms will not be able to respond to changes that rapid, and will simply feel the averaged potential of a slightly shallower trap.

In our system, the timescales line up in a fortuitous manner. The scattering rate of the 841 nm transition is $\Gamma_{841} = 50$ kHz. At these powers and beam sizes, the vertical trap frequency is estimated to be $\omega_{\text{trap}} = 500$ Hz. This gives us a convenient range of $\Gamma_{841} \geq f_{\text{blinking}} \geq \omega_{\text{trap}}$. We implemented a system with $f_{\text{blinking}} = 5$ kHz, with a duty cycle of 80% with the ODT light on (and the MOT light is no longer interacting with the atoms) and 20% with the ODT light off (and the MOT light cooling the atoms). This is shown in a diagram in Figure 4.3. In reality we leave the 841 nm light on during the blinking ODT sequence, but the 1064 nm light shifts the atomic transition far enough away that the 841 nm light becomes so far detuned that it has little to no effect on the atoms.

During the blinking sequence, we use 11 Watts of 1064 nm light in the ODT, which is effectively reduced by the blinking to about 10 Watts, corresponding to a trap depth of 20 μ K. The light is

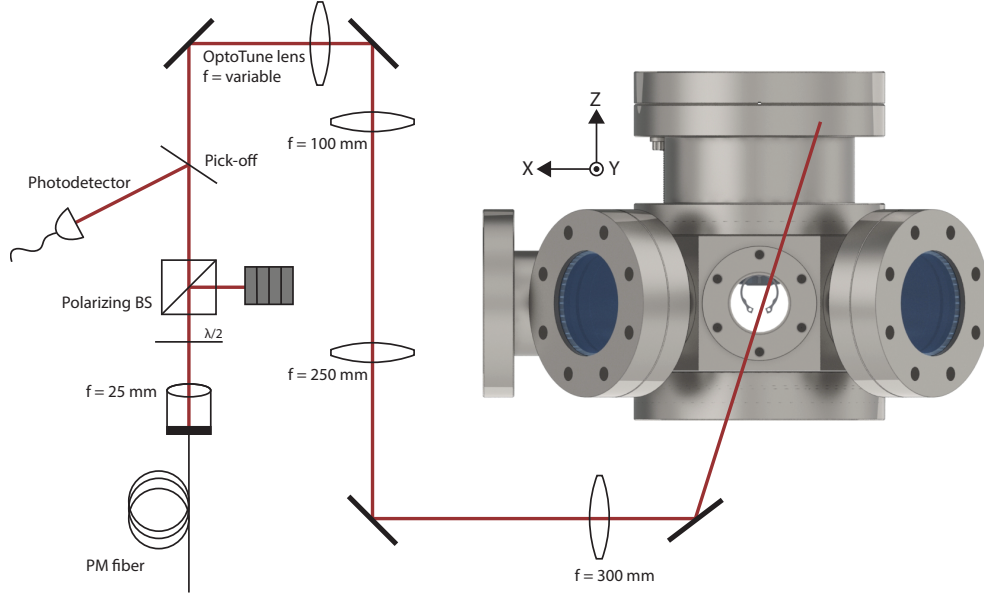


Figure 4.4: Diagram showing the optical layout for the dimple ODT beam path, after the polarization-maintaining Thorlabs fiber. Note that the OptoTune is mounted horizontally. The dimple ODT beam is about 20 deg from gravity.

strobed via an RF switch controlling the RF drive signal of the AOMs controlling the 1064 nm light power into the fiber. The output from the photodetector used for the power stabilization feedback filters out the 5 kHz blinking, enabling the feedback to operate on the averaged power.

This method of strobing the ODT while compressing the narrow-line 841 nm MOT allows us to load 10×10^6 atoms into the ODT at around $2 \mu\text{K}$, in 325 ms, for Er_{168} .

4.2.2 DIMPLE ODT

Once the atoms are captured in the large, horizontal sheet ODT, the small vertical “dimple” ODT beam is also turned on. This is a small beam of 1064 nm light coming from the same laser as the sheet ODT. The AOMs controlling the power of both ODT axes are operating at slightly different frequencies to ensure that no interference effects occur when the two beams overlap. This dimple ODT axis uses up to 1.4 Watts of power on the atoms. Due to the lower power needed in this axis,

we simply use a fiber from Thorlabs to deliver the light to the chamber. This small, vertical beam aligned to the center of the sheet ODT provides us with higher trap frequencies, enabling a more efficient evaporation.

This dimple is designed to give us some amount of independent control over the trap frequencies. In order to achieve this, we use a tunable lens from OptoTune (Part number: EL-16-40-TC (20D)) See Figure 4.4 for the optics layout of the dimple ODT beam. There is a collimation lens after the fiber, then a telescope expanding the beam, then a final focusing lens (with $f=300$ mm) giving us the tight focus on the atoms. The tunable lens is placed before the telescope, in the Fourier plane of the final focusing lens. The OptoTune lens is mounted horizontally so as to avoid deformations of the lens surface due to gravity. In this orientation, gravity would cause a uniform sag of the surface, rather than a deformation that would affect the lens performance.

The default state of the lens surface is flat, and by applying up to ± 250 mA of current, the surface curves into a plano-concave or plano-convex lens. The current range provides a focal length range of -10 diopters to $+10$ diopters. Since the tunable lens is in the Fourier plane of final focusing lens, adjusting its focal length changes the location of the final focus, which changes the size of the beam in the plane of the sheet ODT. We use a current range of up to 40 mA, giving us a range of dimple sizes in the plane of $25\mu\text{m}$ to about $200\mu\text{m}$. For the remainder of this discussion, the size of the dimple ODT in the plane of the sheet ODT will be referred to as the dimple ODT's waist, for simplicity.

To load the atoms from the large, sheet ODT into the additional confinement of the dimple ODT, we turn the dimple beam on at 1.4 Watts with a waist size of $60\mu\text{m}$. At this point in the sequence, the sheet ODT power is at 10 Watts, giving a trap depth of $20\mu\text{K}$. In order to enhance the loading, the sheet ODT's power is then reduced over 200 ms to 7 W, to a trap depth of $14\mu\text{K}$. A single pulse of 841 nm light is also applied to the atoms during this ramp, as described in the next section.

4.2.3 NARROW-LINE COOLING AND EVAPORATION IN ODT

The loading into the dimple ODT was found to be improved by a beam of 841 nm light, acting as an optical molasses and cooling the atoms. This light is turned on at the same time as the dimple ODT, and lasts for 200 ms, during the sheet ODT's initial ramp down in power.

The 841 nm light is slightly red-detuned from the shifted transition of the atoms in the sheet ODT. The absolute frequency of the light of the molasses pulse is kept constant, while the depth of the ODT is reduced. This pulse addresses mainly the atoms that in the sheet ODT but have not yet been successfully loaded into the dimple ODT. The atoms in the dimple are further shifted from the resonance and do not interact significantly with this pulse of light.

At the end of the 200 ms of the 841 nm narrow-line cooling light and the sheet ODT's initial power ramp-down, there remain trapped a total of 5×10^6 atoms. The atom number in the dimple is about 1×10^6 with a phase-space-density of about 0.07. At this point, the 841 nm light is turned off, and only a short evaporation remains to reach degeneracy.

During the evaporation, we follow a standard procedure of ramping down the power in both ODT axes, as shown in Figure 4.1. In addition, we also reduce the size of the dimple beam via the tunable lens, from 65 μm to 32 μm , during the first half of the evaporation sequence. The dimple size is then kept constant for the remainder of the cycle. This allows us to control the trap frequency and the collision rate in the trap, which is crucial for maintaining efficient evaporation sequences for each of the different isotopes of erbium. Each of the isotopes of erbium has a different scattering length, as measured in [28]. For the bosons, they range from 78 a_0 for ^{166}Er to 223 a_0 for ^{170}Er . For ^{166}Er , the isotope used here, the scattering length is 200 a_0 .

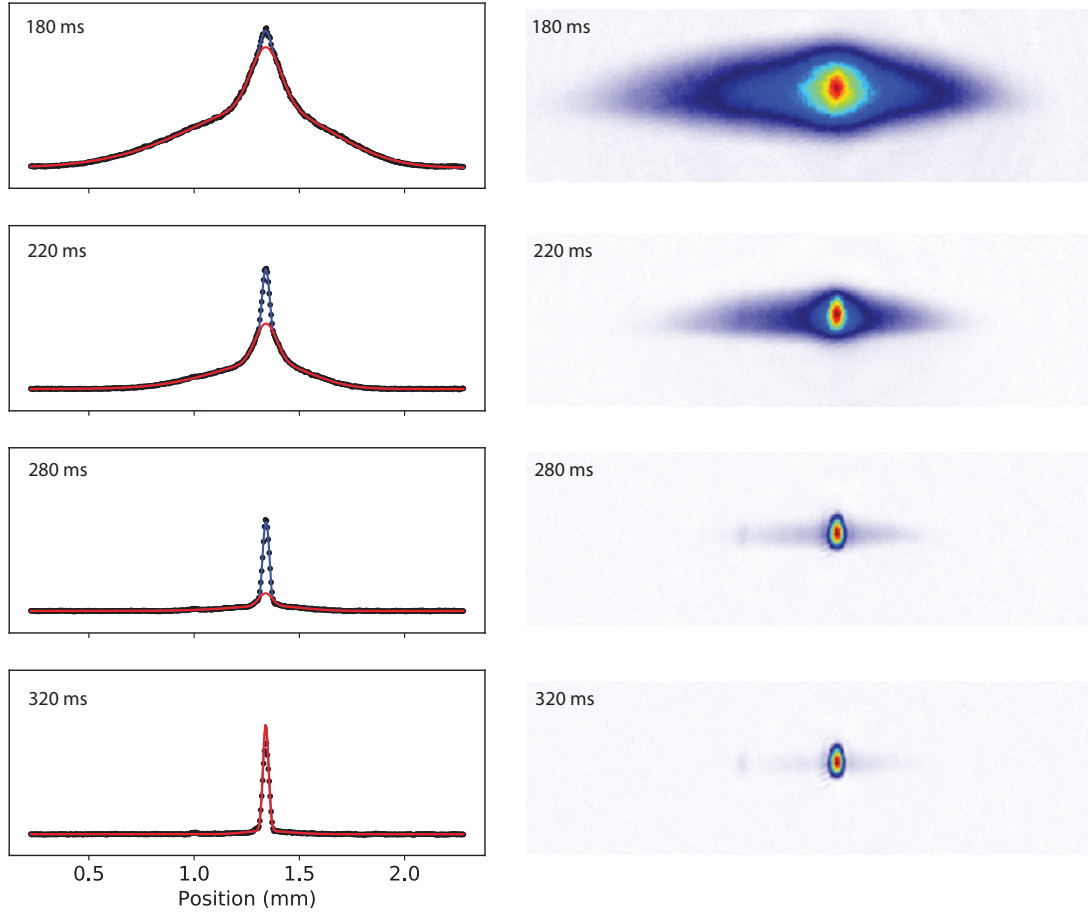


Figure 4.5: Integrated profiles (left) of the absorption images (right) of the atom cloud as the evaporation sequence progresses. By 320 ms into the evaporation, the BEC is nearly pure. To the full data (black dots), we fit an inverted parabola profile summed with two separate Gaussian (blue line), as described in more detail in the text. Excluding the BEC portion of the data, we then fit the sum of two Gaussian functions (red line).

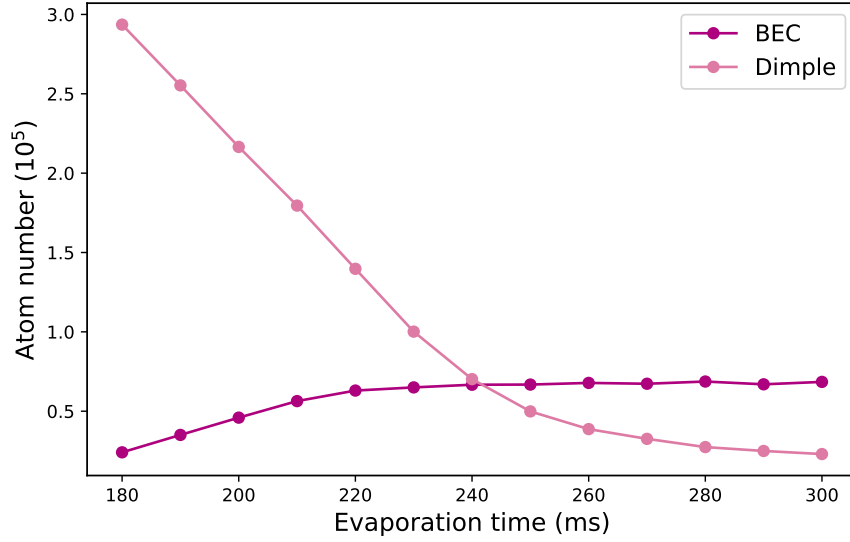


Figure 4.6: Plotted here are the atom number in the dimple (thermal atoms, not condensed into the BEC) and the atom number in the BEC. These are extracted from fits to the integrated profiles shown in Figure 4.5. The atom number in the BEC saturates quickly, after about 225 ms, however the continued evaporation serves to eliminate the thermal dimple atoms.

4.3 SUB-SECOND BEC RESULTS

Very quickly into the evaporation ramp, we begin to see the onset of degeneracy. This is shown in Figure 4.5, from absorption images of the Bose-Einstein condensate after 22 ms of time-off-flight, as well as in integrated profiles of these images. The BEC onset begins after about 180 ms of evaporation, reaching a nearly pure condensate after 320 ms. This is a nearly pure BEC of 80×10^3 atoms after a total time of 850 ms.

The BEC onset is quantified in Figure 4.6. To the profiles shown in Figure 4.5, we fit three separate functions. These describe the BEC with an inverted parabola profile, and the dimple ODT and sheet ODT each with a separate Gaussian profiles. This allows us to extract separately the atom numbers in the sheet ODT, the dimple ODT, and the atom number that has condensed into the BEC. Details of the fit are in the supplementary material for this publication [53]. By fitting pro-

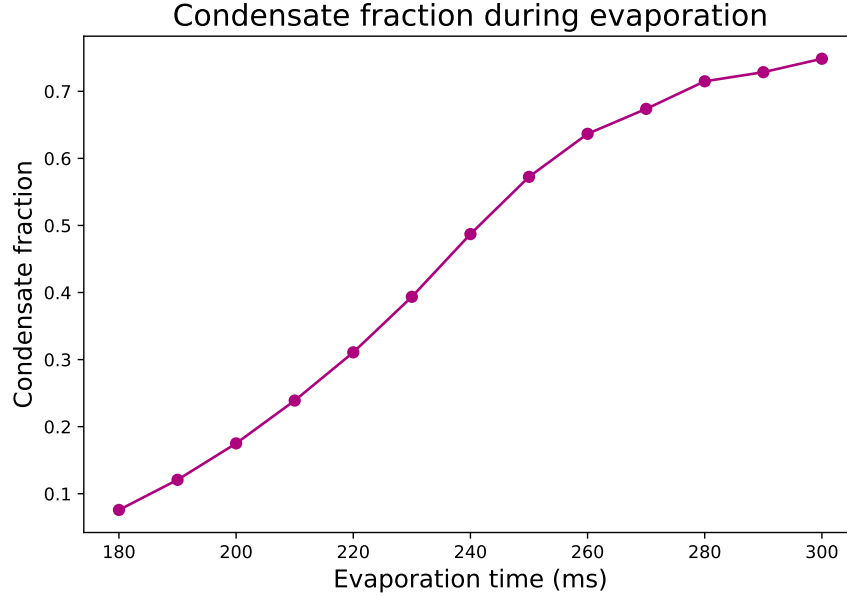


Figure 4.7: Plotted here is the fraction of the total dimple atoms that are condensed in the BEC. That is, the ratio of the atoms in the BEC to the total number of atoms in the dimple (thermal + BEC). This fraction continues to increase throughout the evaporation, seeming to saturate only at the end of the evaporation, after about 300 ms.

files to the full range of data, we obtain a fit capturing the characteristic, inverted parabola shape of the BEC (blue line). We then also fit to the data, excluding the range in which the BEC appears, to emphasize the difference between the Gaussian shape of the thermal cloud (red line).

The number of atoms in the BEC does not increase significantly during the second half of the evaporation after BEC onset. However the end of the evaporation serves to eliminate the dimple atoms that haven't been condensed into the BEC, thus increasing the condensate fraction, which is shown in Figure 4.7. This is the ratio of the atoms that have condensed into the BEC to the total number of atoms in the dimple.

It is also interesting to examine the temperature of the atoms in the dimple over the course of the evaporation. This is shown in Figure 4.8. The temperature drops most significantly in the early half of the evaporation, down about 150 nK after the first 200 ms of evaporation.

The BEC onset temperature T_c is related to the atom number N_c and the geometric trap frequency $\bar{\omega} = (\omega_x \omega_y \omega_z)^{1/3}$ by the following expression, from [56]:

$$N_c = g_3(1) \left(\frac{k_b T_c}{\hbar \bar{\omega}} \right)^3$$

At 180 ms into the evaporation ramp, with a measured 3×10^5 atoms in the dimple, and with trap frequencies estimated at about $\omega = (50, 50150)$ Hz, we estimate the BEC onset temperature to be 210 nK. This is consistent with what we observe. The onset of a measurable condensate fraction, as shown in Figure 4.7, happens right when the temperature reaches about 200 nK, as shown in Figure 4.8.

4.4 DEGENERATE FERMI GAS

In order to create degenerate Fermi gases (DFGs) with the fermionic isotope ^{167}Er , we use a series of ramps that are qualitatively similar to the ramps used for the fast BEC creation of ^{168}Er . We were able to create DFGs with 1.5×10^6 atoms at $T/T_F \approx 0.25$ in a total cycle time of about 5 s. This is significantly slower than the sub-second cycle time achieved for the BECs, however, to our knowledge, it is still a substantial improvement relative to the standard in the field for the sample preparation time for degenerate Fermi gases.

Without long-range interactions, fermions do not collide at low temperatures [57]. This means that the standard evaporation procedures used to cool bosons to degeneracy cannot be used to cool fermions beyond a certain temperature, once their scattering cross section has vanished. However, for dipolar atoms, the dipole-dipole interaction provides a long-range interaction which contributes to the scattering cross section, allowing it to remain non-zero at very low temperatures. This is discussed in more detail, and shown experimentally in dipolar, fermionic ^{161}Dy [58]. The first efficient, direct evaporation of a fermionic atom deep into degeneracy is shown with ^{167}Er in [22].

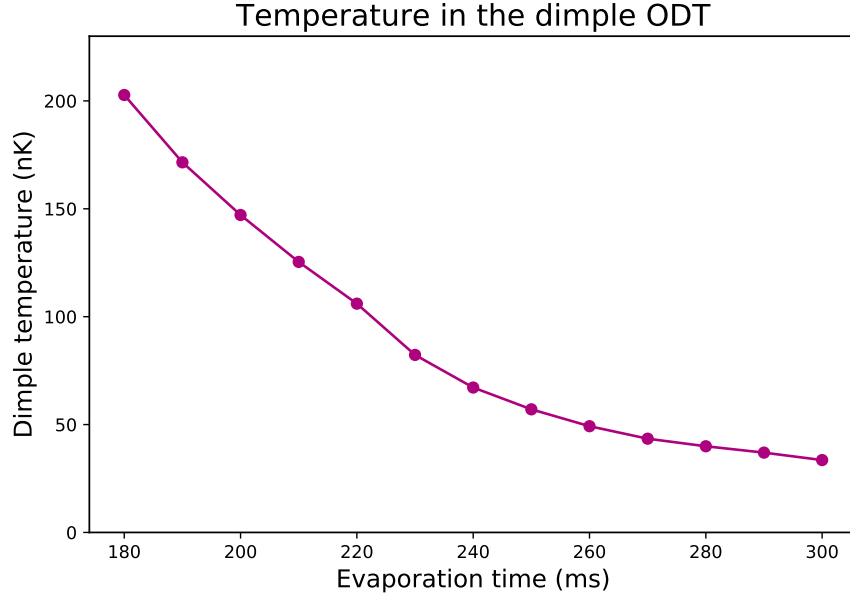


Figure 4.8: Plotted here is the temperature of the atoms in the dimple as a function of the evaporation time.

In our work, we show an experimental sequence for creating erbium DFGs that is only different from the BEC sequence in the details of the ramp parameters. The fermionic MOT loads in a longer time due to the reduced natural abundance of the fermions due to the population in several of the hyperfine states. Additionally, the evaporation is significantly slower than for ^{168}Er , due to the reduced effective scattering length of the fermions relative to ^{168}Er [22]. We use a dimple size of about $250\text{ }\mu\text{m}$, much larger than for the bosons, in order to maintain the ratio of elastic to inelastic collisions.

Figure 4.9 shows an absorption image of an erbium DFG. The image is taken after 22 ms of time-of-flight. The integrated profiles along the vertical and horizontal axes of the image are also shown. A function formed by the sum of a poly-logarithmic function plus a Gaussian are fitted to each of the integrated profiles. The fugacity $e^{\mu/k_B T}$ is extracted from the fits in each axis, and from this fugacity we can calculate the temperature of the atoms. The resulting temperatures are $T/T_F = 0.22(2)$ in the vertical and $T/T_F = 0.24(2)$ in the horizontal.

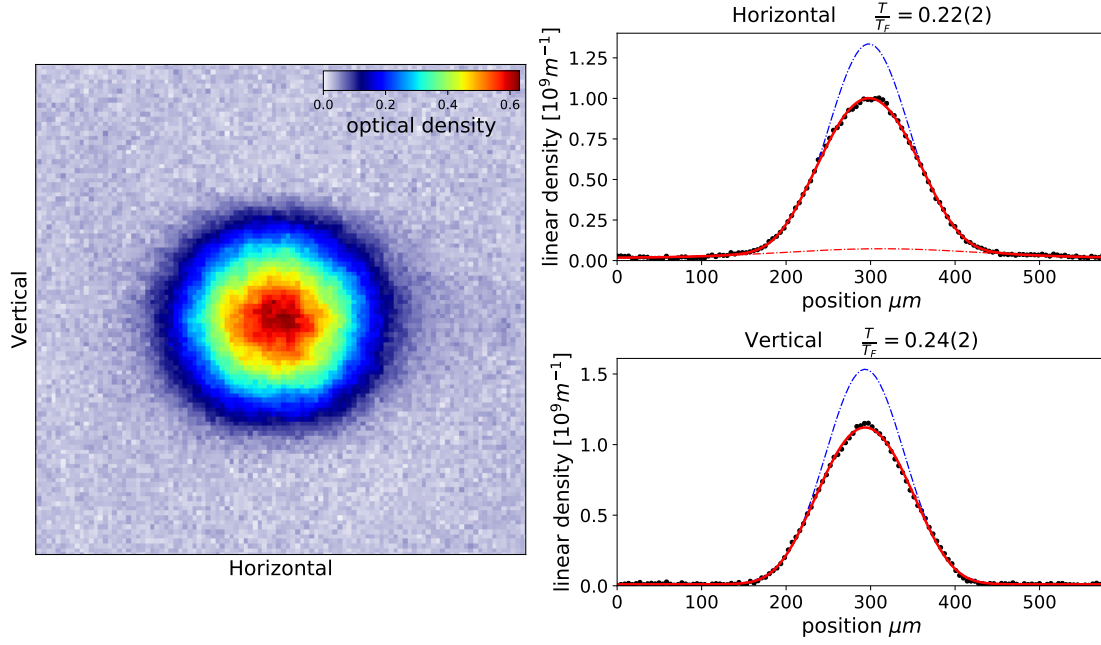


Figure 4.9: Degenerate Fermi gas: right, an absorption image of a single measurement of a DFG of 130,000 atoms of ^{167}Er , taken after a 22 ms time-of-flight. Left, the integrated profiles along the horizontal (upper image) and vertical (lower image) axes. A function that is the sum of a poly-logarithm and a Gaussian (solid red line) is fitted to the measured integrated distributions (circles). We can extract the temperature of the cloud along each axis, obtaining for the vertical axis $T/T_F = 0.22(2)$ and for the horizontal axis $T/T_F = 0.24(2)$. To emphasize the deviation from a classical gas, we also fit a Gaussian (dashed black lines) to the integrated profiles. Figure from [53].

5

Magnetic field control

Magnetic fields are an extremely important tool for manipulating atoms in ultracold atom experiments. Here, they are crucial in the performance of the spin-flip Zeeman slower, where a spatially varying magnetic field shifts the atomic energy levels via the Zeeman effect. Magnetic field gradients give us a spin-dependent way of interacting with the atoms, leading to magneto-optical traps and spin-dependent imaging of the atoms. Magnetic field offsets in the main chamber allow us to control the orientation of the atom dipoles, and, via Feshbach resonances [59], allow us to tune the

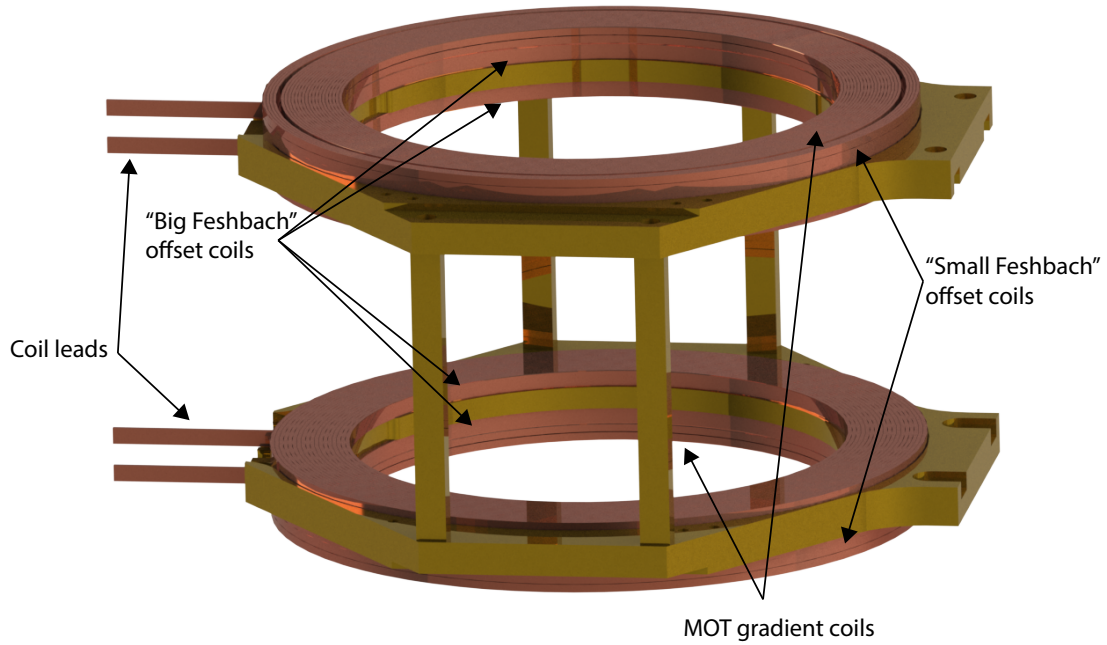


Figure 5.1: 3D rendering of the assembly of the offset and gradient magnetic field coils around the main chamber, including the mounting and the cooling plates.

interaction strength between the atoms.

In this chapter, I describe the coils in the experiment that allow us to control the atoms in the ways mentioned above.

5.1 MOT AND FESHBACH COILS

In order to control the vertical magnetic field offset and gradient at the position of the atoms once they are trapped, we have an assembly of coils mounted to the main chamber, centered around the atom location. This assembly of coils around the main chamber is shown in Figure 5.1. The coil assembly is formed of three sets of coils: the “big Feshbach” offset coils provide large magnetic field offsets of up to 500 Gauss, the “small Feshbach” offset coils provide small, rapid offsets of up to 5

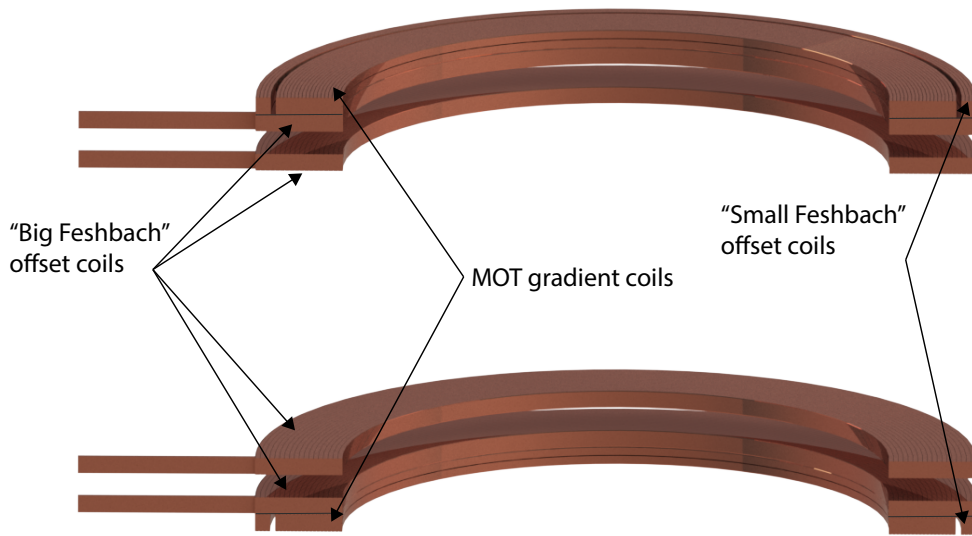


Figure 5.2: Cross-section of the MOT and Feshbach coils assembly, without the cooling plate and mounting. This view shows clearly the three different coils that make up each half of the coil assembly.

Gauss, and finally, the MOT gradient coils provide the gradient for the MOTs, of up to about 10 Gauss/cm. The total coil assembly consists of two identical sets of coils, one above the main chamber and one below the chamber. The coils are symmetric about the x-y plane in the center of the chamber, as shown in Figure 5.3.

The big Feshbach coil set is designed to provide very large offset magnetic fields, of up to 500 Gauss, in order to explore the high-field Feshbach spectrum of erbium. This spectrum has been mapped out to 70 Gauss [59], so this design gives us the possibility of going to even higher fields. In order to achieve this high field, we need many turns of wire in the coils, and we need a wire that can handle high currents. This coil pair consists of two coils, one above and one below the chamber. Each coil is made up of two layers of 18 turns each, as shown in Figure 5.2. The two layers are separated by a water-cooled plate, which is described in more detail below. To maximize the uniformity of this magnetic field, the big Feshbach coils are made in a Helmholtz configuration.

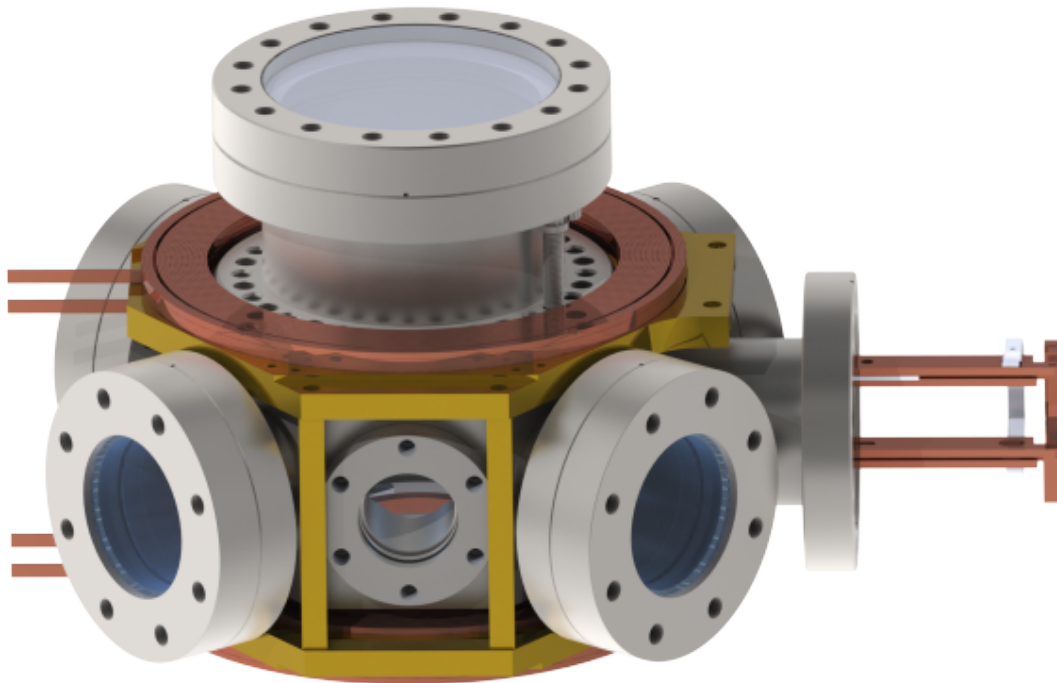


Figure 5.3: Side view of the main chamber, showing how the Feshbach and gradient coil assembly are mounted around the chamber.

Since the “big Feshbach” offset coils are estimated to dissipate up to almost 3 kWatts of power at the maximum field they’re designed for, we added a cooling plate in between the two layers of each of the coils of the pair. The cooling plate is shown within the coil assembly in the diagram in Figure 5.1. The cooling plates consist of a brass plate, with two turns of $\frac{3}{8}$ ” 122 copper tubing press-fit into machined grooves. The plate can be seen in more detail in Figure 5.4.

The “small Feshbach” offset coil pair is designed to provide small adjustments to the offset magnetic field of up to about 5 Gauss. This coil pair has fewer turns, 5 turns per coil, which reduces the inductance and allows this coil to respond more rapidly than the big Feshbach coils. In order to again minimize gradients caused by this set of offset coils, they are placed in nearly Helmholtz configuration, although physical constraints prevent them from being exactly Helmholtz.

The MOT gradient coil pair is designed to provide the magnetic field gradient to operate the MOTs, of up to 10 Gauss/cm. Each coil in this pair has 20 turns of wire. To create as pure of a magnetic gradient as possible, these coils are made in a nearly anti-Helmholtz configuration, although once again physical constraints due to the presence of the big Feshbach coils and the chamber prevent this from being a perfect anti-Helmholtz alignment.

The MOT coils are driven with a 200 W power supply delivering 0-10 V and 0-20 A (TDK-Lambda Z10-20-LAN-U). The small Feshbach coils are driven with a 48 W power supply delivering 0-8 V and 0-6 A (Hewlett Packard E3614A 48W). For big Feshbach coils, we purchased a pair of 3300 W power supplies, delivering 0-33 V and 0-110 A (Delta Elektronika SM66-AR-110). These supplies showed a significant amount of instability, shot to shot, which cause the position of the field zero to fluctuate. This meant the final position of the MOT was fluctuating, and affected the loading into the ODT. For this reason, we are currently operating with a smaller, more stable supply (Hewlett Packard E3614A 48W) and will need to install some current stabilization on the larger supplies before going to higher field.

5.1.1 CONSTRUCTION DETAILS

The coil assembly is made by first winding the “big Feshbach” coil, epoxying it to the cooling plate. The next step is winding the two sets of smaller coils, the “small Feshbach” coil and the gradient coil, and in turn epoxying them to the “big Feshbach” coil. The steps and materials used are detailed below.

The big Feshbach coil is first wound around a home-built Plexiglas jig which constrains the dimensions desired for the coil. As shown in Figure 5.2, each of the two coils in big Feshbach pair is formed of two layers, in between which sits the cooling plate. The jig is made of circular layers of Plexiglas or varying radii. This ensure the coil is clamped on the outside, constrains the inner radius of the coil, and provides a spacer to mimic the cooling plate’s thickness. In this way, we can easily

wind the coil to the exact dimensions required, secure the wire (we used zip-ties), then remove the coil from the jig and place it around the cooling plate.

The wire used has a rectangular cross-section of 1.35 x 5.11 mm, made by Sam Dong America. It is very stiff, hence the need for carefully-designed jigs to help with the winding. After winding the two 18-turn “pancake” layers, the coil is held in place by many zip-ties. We then removed it from the jig and placed it around the cooling plate.

Once the cooling plate is placed between the coil layers, we cannot simply epoxy the whole coil at once, because the wire starts to unwind as soon as the zip-ties are removed. To circumvent this, we simply glue a small section of the coil, while the rest of the coil is still well-secured by the zip-ties. We isolated the coil from the cooling plate with some parchment paper to ensure that they aren’t glued together yet at this stage. During the epoxy curing stages, we used an aluminum plate on both sides of the assembly to clamp the coil together tightly to ensure it stays flat and as well-contacted to the cooling plate as possible. After the first section of the coil has been glued, we removed all the zip-ties and applied epoxy to the entire coil, including to the cooling plate. We applied the epoxy on both sides of each layer of the coil, rubbing it in to ensure it permeates all the way through each layer. At this point in the process, we installed a few thermocouples that we want to have to monitor the temperature of these coils. We then clamped the assembly between sheets of parchment paper and the aluminum plates, and cured the whole thing in a toaster oven at 250 F for an hour, and then at 350 F for another hour, as directed by the epoxy manufacturer.

The gradient coils and the “small Feshbach” coils are made from slightly smaller rectangular wire, with a cross-section of 0.94 x 2.62 mm, made by HSM Wire International. We first wound the gradient coil in another Plexiglas jig. This is a single layer with 20 turns, which means the jig is simpler. We added two slots to the jog, permitting us to epoxy two sections of the coil while it’s still in the winding jig, meaning it’s also still being clamped and held flat during the first epoxy stage. After the first sections of epoxy are cured (about 24 hours at room temperature) we remove the coil from the

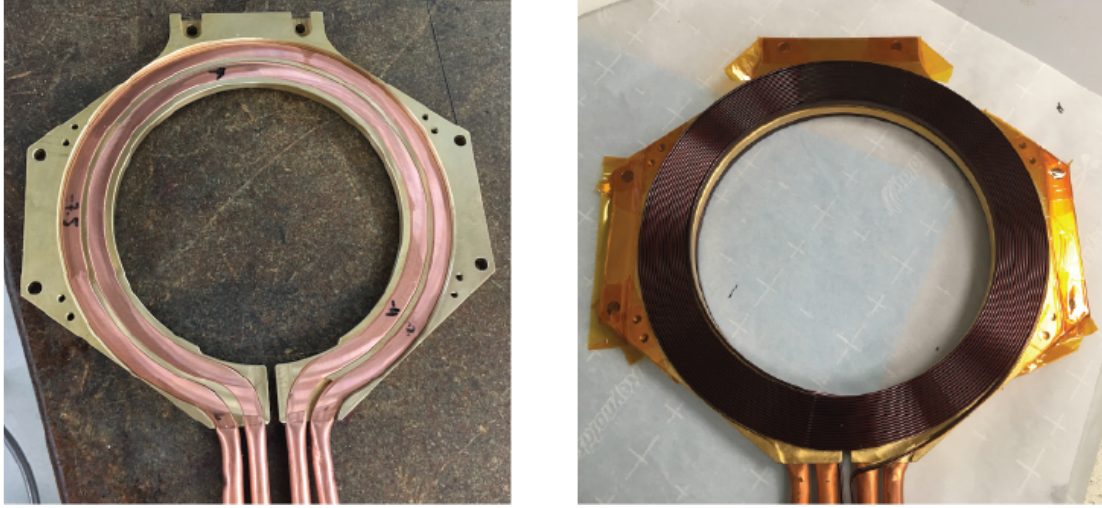


Figure 5.4: Left: picture of the machined cooling plate, with the copper tubing visible. Right: picture of the Feshbach coil epoxied onto the cooling plate.

jig.

We used vacuum impregnation to epoxy the remainder of the coil. We placed the coil in a home-built “vacuum chamber” (made from a thick plastic mixing bowl, a 1” thick sheet of Plexiglas, a rubber seal, a few valves and connectors, and an oil-sealed roughing pump) and submerged it with about 300 g of epoxy. When we pump the chamber out, this removes all the air bubbles that are in the epoxy, thus improving the strength and thermal conductivity of the assembly. We left the coil in the rough vacuum for about 15 minutes, after which we saw fewer bubbles coming up out of the epoxy on the coil. We took the coil and wiped excess epoxy very carefully, and then proceeded to carefully secure the leads with zip-ties, and cured the coil in the toaster oven.

For the “small Feshback” coil, the last one of the assembly, we simply wind it directly around the gradient coil, using the same winding jig. We used the same rectangular wire for the “small Feshback” coil as for the gradient coil, so the transition between the two is very smooth. For the “small Feshback” coil we wound 5 turns. As before, we glued a small section of the coil together and to the gradient coil, then removed it from the jig and finished applying epoxy to the rest of the coil, and

then cured it in the toaster oven.

We then epoxied the “small Feshbach” and gradient coil assembly to the “big Feshbach” and cooling plate assembly. This required a little bit of sanding and shaving of epoxy in some places to optimize how everything fit together. We also added two more thermocouples to the assembly at this stage, before curing it in the toaster oven for the usual hour at 250 F and another hour at 350 F.

The epoxy used for the entire assembly process described here is Duralco 4525, from Cotronics. It has a thermal conductivity of 1.87 W/mK and a low shrinkage. It worked very well for us, despite its slightly intimidating, gooey, black appearance. And at the time of writing the thesis, the coils have been assembled for almost 4 years with no sign of any kind of deterioration.

5.1.2 COIL TESTS

We mounted the coils in the same way as they will be assembled in the final experiment setup, and measurement the magnetic field created by each coil. We scanned the position of the magnetic field probe in the horizontal $x - y$ plane of the atom beam, from the coil centers to about 75 mm away in both directions. The resulting magnetic field measurements are in Figure 5.5. The measurements were done applying a 25 Hz oscillating current to the coils. We measured the amplitude of the resulting oscillating magnetic field, which allowed us to ensure we had no background offsets or stray signals.

From these measurements we were able to extract the transfer function of the coils, i.e. the resulting field/current for the offset coils, and the gradient/current for the gradient coils. We obtained a field for current of 3.21 Gauss/Amp for the big Feshbach coils, a field for current of 0.748 Gauss/Amp for the small Feshbach coils, and an axial gradient for current of 0.493 Gauss/cm/Amp for the MOT coils.

We also are able to extract the relative centration of the three different coil pairs from these mea-

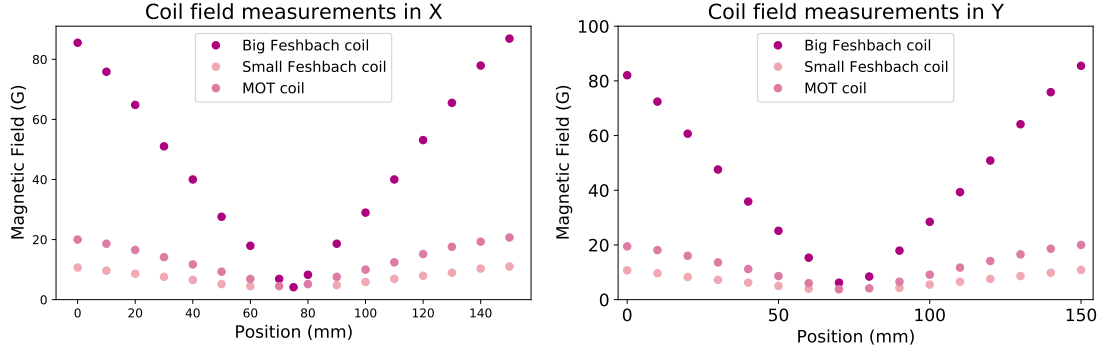


Figure 5.5: Measured magnetic field from the big Feshbach, little Feshbach, and gradient coils in the main chamber coil assembly. Y is the direction of the atom beam (+Y is in the direction of the atom velocity), and X is perpendicular to the atom beam.

measurements. We fit the following equation to the data

$$B(x) = \sqrt{(m(x - x_0))^2 + c^2}$$

where m is the gradient in G/mm, x_0 is the position offset in mm, and c is any background offset or sensor offset in G. Fitting to the measured field for the three coils and along both axes, we estimate that the coils are centered relative to each other up to 2 mm.

With the coil assembly mounted around the main chamber (made of 316 stainless steel), we measured the response of the gradient coils to fast changes in the drive current. We wanted to estimate how quickly we are able to change the magnetic field gradient on the atoms. We found that the turn-on time is about 10 ms. In order to change the gradient more rapidly than this timescale, we will have to use coils with lower inductance and inside the chamber. See Section 5.5 for details on our in-vacuum coil.

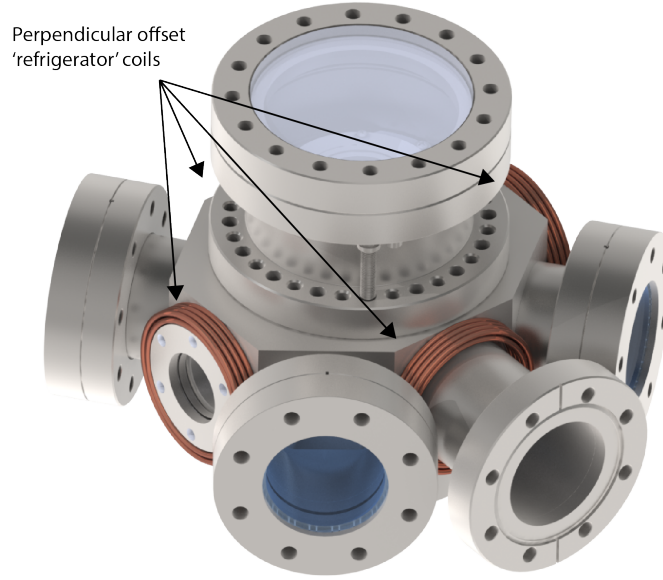


Figure 5.6: Rendering of the main chamber showing the location of the 4 offset coils. The main coil assembly is removed from the render to increase visibility of the offset coils.

5.2 PERPENDICULAR “REFRIGERATOR” OFFSET COILS

The coil assembly described in the previous section provides the option of applying on the atoms a magnetic field gradient, and a vertical magnetic field offset in the $\pm Z$ (gravity) direction. We seek to have full control over the direction of a magnetic field offset on the atoms, and to be able to rotate the field direction. For this reason, we have an additional set of small coils which allow us to create magnetic fields along the X and Y directions. The placement of the coils on the main chamber is shown in Figure 5.6. The coils producing a horizontal magnetic field along the atom beam direction (which allows us to cancel stray fields from the Zeeman slower during the MOT loading) are fitted around the flanges connecting the Zeeman slower and the dump chamber adapter to the main chamber. The coils producing a horizontal magnetic field perpendicular to the atom beam direction are fitted around the viewports, as shown in the figure.

These “refrigerator” coils are wound from hollow copper tubing, similar to the tubing used in refrigerators, hence the name. The coils are 8 turns total, 2 layers of 4 turns each. We run water directly through the hollow copper wire in order to cool the assembly.

The set of coils mounted on the viewports, generating a field perpendicular to the atom beam, are driven by a TDK-Lambda power supply (GEN 12.5-120), generating 0-12.5 V and 0-120 A. The set generating a field along the atom beam is driven by a Delta Elektronika supply (SM 7.5-80), generating 0-7.5 V and 0-80 A.

5.3 ZEEMAN SLOWER AND ADDED COILS

In order to slow the atoms to the capture velocity of the MOT (about 5 m/s for our first MOT), we use a Zeeman slower. A beam of light, counter-propagating with the atom beam, slows the atoms down as they absorb and re-scatter photons from this beam, addressing the atoms on the 401 nm transition. Typically, atoms start out traveling very quickly, about 1000 m/s, and are slowed down to 5-10 m/s (in our case). The initial velocity is large enough that the atoms see a significant Doppler-shift, causing the light to no longer be resonant with the atomic transition once the atoms have been partially slowed. This shift prevents the atoms from being further slowed by the light. In order to compensate for this, the Zeeman slower coils are used to apply a magnetic field that varies spatially along the atom beam over the length of the Zeeman slower, ending in the main chamber. This varying magnetic field causes a varying shift in the energy levels of the atoms, which can be tuned to exactly compensate for the Doppler-shift.

The magnetic field profile $B(x)$ required to ensure the Doppler-shift is compensated is one that satisfies the following condition:

$$\mu B(x) - \hbar k v(x) = \hbar(\omega - \omega_0)$$

Here, μ is the difference in the magnetic moment of the ground and the excited state, $v(x)$ is the velocity profile of the atoms, and $\omega - \omega_0$ is the detuning of the laser light from the resonance. Solving for the magnetic field profile, assuming a constant acceleration and hence a square-root profile for the velocity, gives us:

$$B(x) = B_0 \sqrt{1 - x/x_0} + B_{\text{offset}}$$

Where B_{offset} is a constant offset, and the length of the Zeeman slower is $x_0 = (\Delta v)^2/2a$. Here (Δv) is the change in velocity over the course of the Zeeman slower, and a is the acceleration. This acceleration is limited to $a_{\text{max}} = \hbar k \Gamma / 2m$, by the force a single photon can exert on an atom on this transition.

In collaboration with the Ferlino group in Innsbruck, we used their design for the Zeeman slower coils, which is outlined in [28]. The design consists of six independent coils, giving a lot of tunability in the exact profile by tweaking the individual current values. The coils are wound around a tube which has an isolated outer layer for water-cooling the coils, around the UHV section.

We have also added a set of three more separate coils, wound around the main chamber end of the Zeeman slower. These coils are wound from the “refrigerator wire” described above for the offset coils. Since these coils are directly cooled from the inside, the cooling is more efficient and does not heat up the main chamber. In comparison, the Zeeman slower coils, under continuous operation, were found to heat the chamber up by up to 30° C.

These new coils were simply wound around the Zeeman slower once the chamber was already assembled. They are held in place by 3D printed clamps, which are screwed around the Zeeman slower.

5.4 EARTH COMPENSATION COILS

In order to cancel out Earth's field and other stray fields from the building or neighboring labs, we built a set of 3 large, perpendicular coil pairs around the entire experiment's optical table. The coils were wound directly into U-channels that are mounted on the 80-20 cage around the experiment table, and zip-tied into place.

The coils on the two small ends of the optical table, perpendicular to the atom beam direction, are made from 10 AWG wire. The coil dimensions are roughly 1.52×1.56 m, and the two coils in the pair are separated by about 3.27 m. Each coil in the pair is made of 42 turns.

The coil pair providing a vertical offset magnetic field, wound in planes perpendicular to the direction of gravity, are made from 14 AWG wire. The coil dimensions are roughly 3.28×1.73 m, and the two coils in the pair are separated by about 1.59 m. Each coil is made of 25 turns.

The coils on the long sides of the optical table, are made from 14 AWG wire. The coil dimensions are roughly 3.19×1.51 m, and the two coils in the pair are separated by about 1.73 m. Each coil is made of 25 turns.

The 6 coils are each driven by an independent power supply, although for now we have always been running them with the same current going through both coils of each pair. They are all powered by TDK-Lambda Z-20-10 linear supplies. The resulting magnetic field from these 3 coil pairs is estimated theoretically in Figure 5.7, for the approximate current values that we typically run them at.

We typically use spectroscopy signals to find the right values for the compensation fields. By releasing the atoms and letting them undergo a time of flight with all other experiment coils off, we can adjust the currents in the compensation coils using the spectroscopy signals.

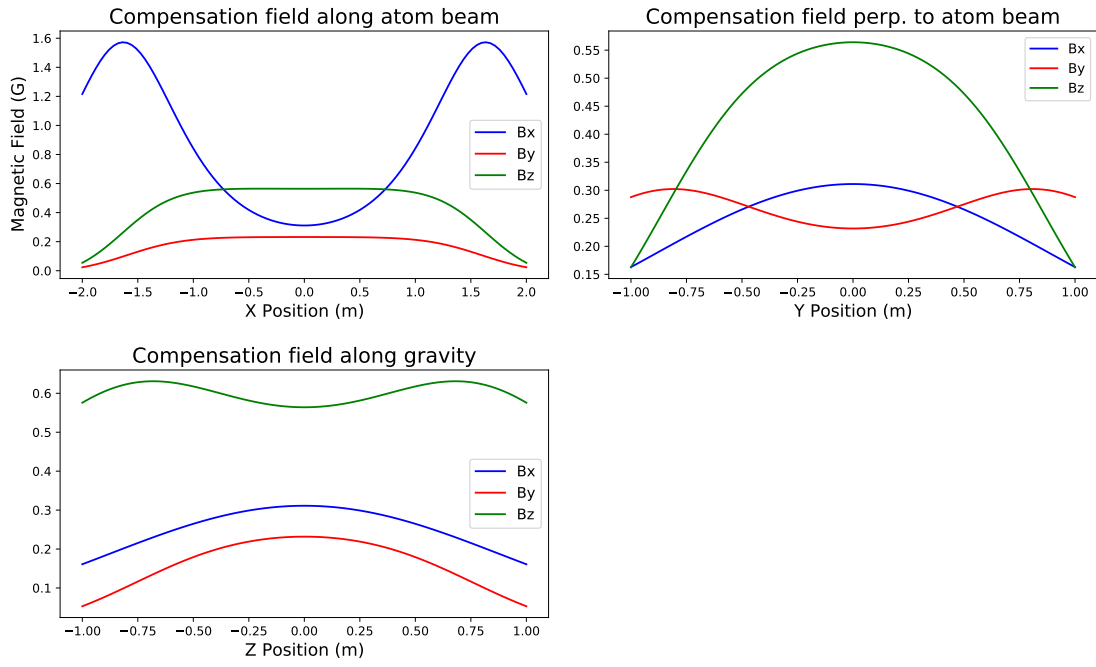


Figure 5.7: The three perpendicular components of the magnetic field created by the compensation coils wound around the entire experiment optical table. Each subplot shows how the three magnetic field components varies along the X, Y and Z directions. X is along the atom beam direction, Y is perpendicular to the atom beam direction, and Z is vertical, along gravity.

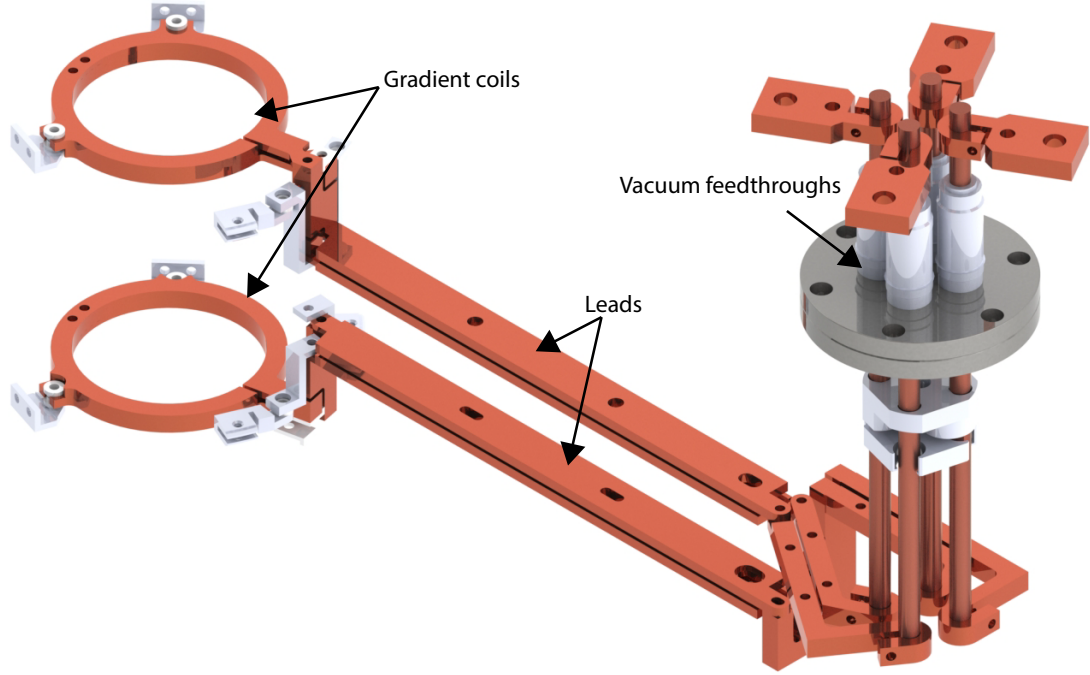


Figure 5.8: Full view of the gradient coil, removed from the vacuum chamber, along with the leads and the vacuum feed-throughs for the electrical connections.

5.5 IN-VACUUM GRADIENT COIL

In order to be able to apply rapid, very large magnetic field gradients on the atoms, given that they are in the center of a large, stainless steel chamber, we implemented an in-vacuum gradient coil. The motivation for this coil is the desire to be able to separate the spin states of the atoms initially trapped in the 2D lattice in order to be able to take spin-resolved images of the atoms.

If we release the atoms from the 2D lattice confinement and apply a large magnetic field gradient the atoms will feel a force

$$F_{\text{magnetic}} = \nabla(\vec{\mu} \cdot \vec{B})$$

Each of the 13 spin states (m-levels) of erbium has a different strength of magnetic moment, and will therefore feel a different force due to this magnetic field gradient, allowing us to use this as a

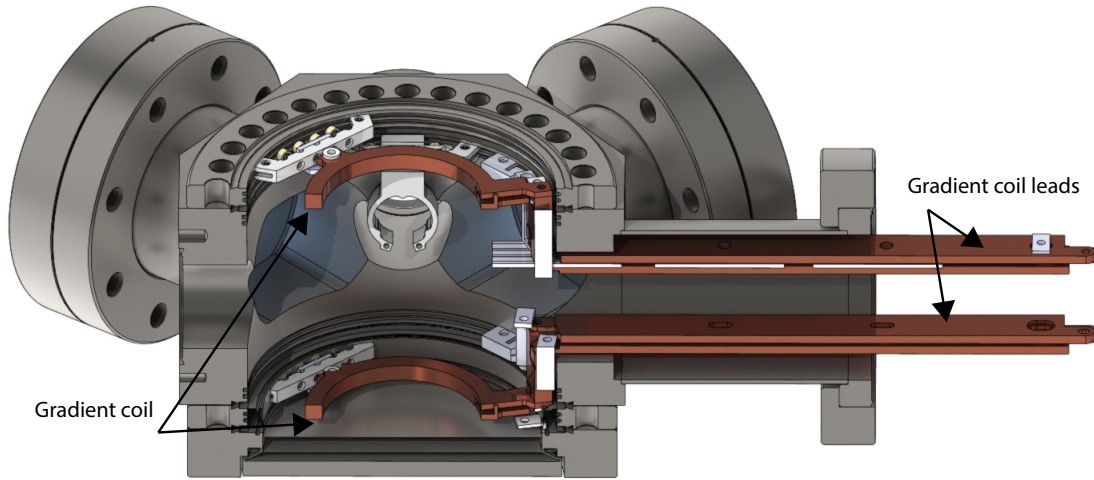


Figure 5.9: Sectioned view of the main chamber, showing the two coils forming the gradient coil pair, in an anti-Helmholtz configuration. The groove grabbers holding the coils to the grooves machined in the main chamber are also shown.

tool to separate them. The momentum imparted to the atoms from this magnetic field gradient is a product of the length of time during which the gradient is applied to the atoms, and the strength of the gradient. However, it is also important for this separation to happen in as short a time as possible, since the wavefunction of the atoms will also be expanding with time as soon as they are released from the potential of the lattice. The broader the wavefunction of the atoms at the time of imaging, the harder it will be to resolve the individual atoms from each other. This means that to achieve the spin separation, and to retain a high imaging fidelity, we need to use a very large, and very short magnetic field gradient pulse.

The goal is to achieve gradients of 100 Gauss/cm. In order to achieve this, we opted to use a single-turn, in-vacuum coil pair, in an anti-Helmholtz configuration. This reduces the inductance, reduces the effects of eddy currents in the steel chamber, and allows us to use smaller coils, placed closer to the atoms. The coils are shown in a section view of the main chamber in Figure 5.9. The complete assembly of the coils, the leads, and the vacuum feedthroughs for the electrical connec-

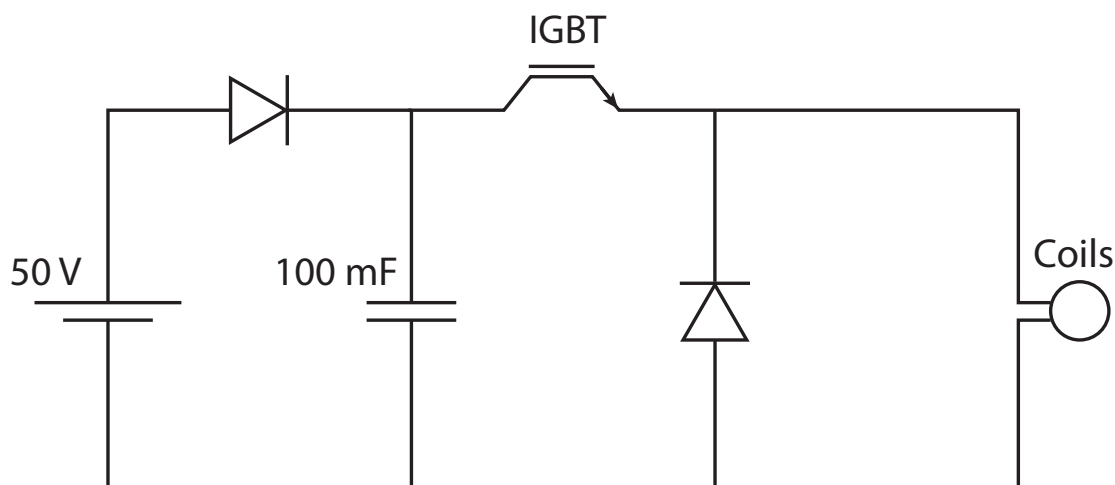


Figure 5.10: Circuit diagram schematic for the in-vacuum gradient coil's driver.

tions are shown in Figure 5.8.

The coils and leads are machined out of copper. The coils are mounted to the inside of the main chamber via groove grabbers, which can be seen in Figure 5.9. Coil is mounted to the groove grabbers via custom macor parts. The two leads going to each coil are isolated from each other with thin plates of macor. The vacuum feedthroughs are in the “dump chamber” section of the vacuum chamber. The leads run out of the main chamber, into the dump chamber, without interfering with the Zeeman beam. All of the hardware used to mount and assemble the gradient coil system is titanium. The coils and driver circuit were designed by post-doctoral researcher Susannah Dickerson.

We estimated that the desired separation of the spin states can be reached by applying 5 kAmps of current through the coils over a few 100 μ s. Since we only need the pulse for such a short duration, we can reach this high current with the help of a capacitor bank. The driver circuit diagram is shown in Figure 5.10. The 100 mF capacitor bank consists of four electrolytic capacitor with a capacitance of 22000 uF each, capable of storing up to 350 Volts of charge. The capacitor part number is United Chemi-Con E37F351CPN223MFM9M. The IGBT driver used in the coil driver circuit is VLA553-01R, able to isolate up to 4000 Volts.

The coils have been used so far to apply magnetic field gradients to the atom cloud to perform Stern-Gerlach measurements of the spin purity of the samples.

6

Accordion Lattice

In certain ultracold atom experiments, it is extremely useful to build an “accordion” optical lattice system, enabling continuous control over the spacing of the optical lattice, while the atoms remain trapped. Such dynamic optical lattices have been implemented as a method of compressing an atom cloud into an effectively “2D” geometry, as in [60], [61], [62], [63], amongst many others.

For our experiment, there are two main applications for such tunable, accordion lattices. The first is our vertical accordion lattice. After reaching quantum degeneracy, the atoms will be loaded

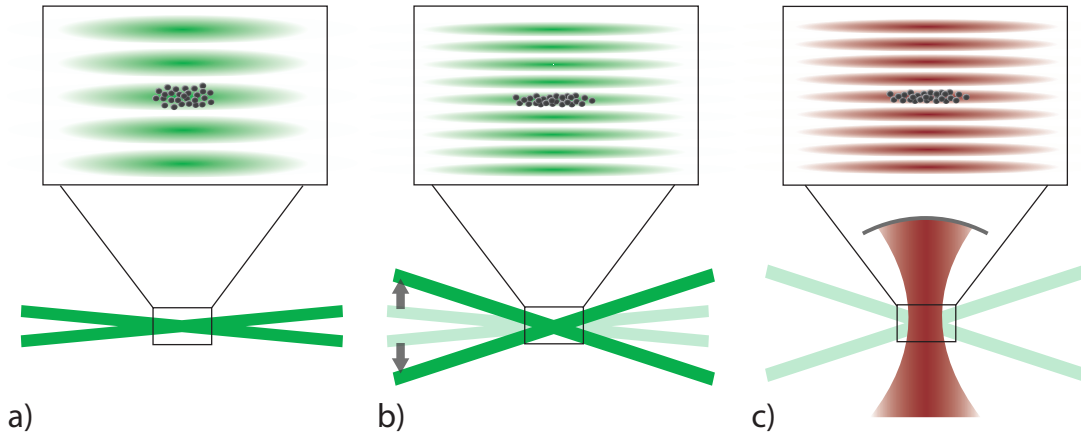


Figure 6.1: Vertical accordion lattice sequence. Atoms are loaded from the optical dipole trap into a wide spacing accordion lattice, shown in a). The angle of the accordion lattice beams is increased, as shown in b). This change in angle of the lattice beams compresses the lattice until the spacing is small enough to transfer the now-compressed atom cloud into the final vertical lattice, as shown in c). The final vertical lattice is a fixed-spacing lattice, simply a beam retro-reflected off of a curved mirror.

from the optical dipole trap into the vertical accordion lattice, which will then vertically compress the quantum gas into a thin layer. This will allow us to load reliably into one, and only one, layer of the final vertical optical lattice which provides confinement within the $x - y$ plane, as shown in Figure 6.1. The second application is for our horizontal accordion lattice. This requirement is due to the fact that we’re studying the dipole-dipole interaction, which has a $1/r^3$ scaling, with high-resolution imaging. The horizontal accordion lattice permits the atoms to be interacting in a lattice with a small lattice spacing, thus maximizing the strength of the dipole-dipole interaction. After interacting, the atoms can then be pulled apart by a 2-dimensional accordion lattice within the $x - y$ plane, until the final spacing is large enough to give a high fidelity for the site-resolved imaging. The atoms are first loaded in the “science” 2-d horizontal lattice, which provides a low-disorder potential in which to perform experiments. Then, the atoms are transferred into the 2-d horizontal accordion lattice, with a small initial spacing, which is then expanded as in Figure 6.2.

To achieve this continuously tunable lattice spacing, we use the fact that the spacing s of an opti-

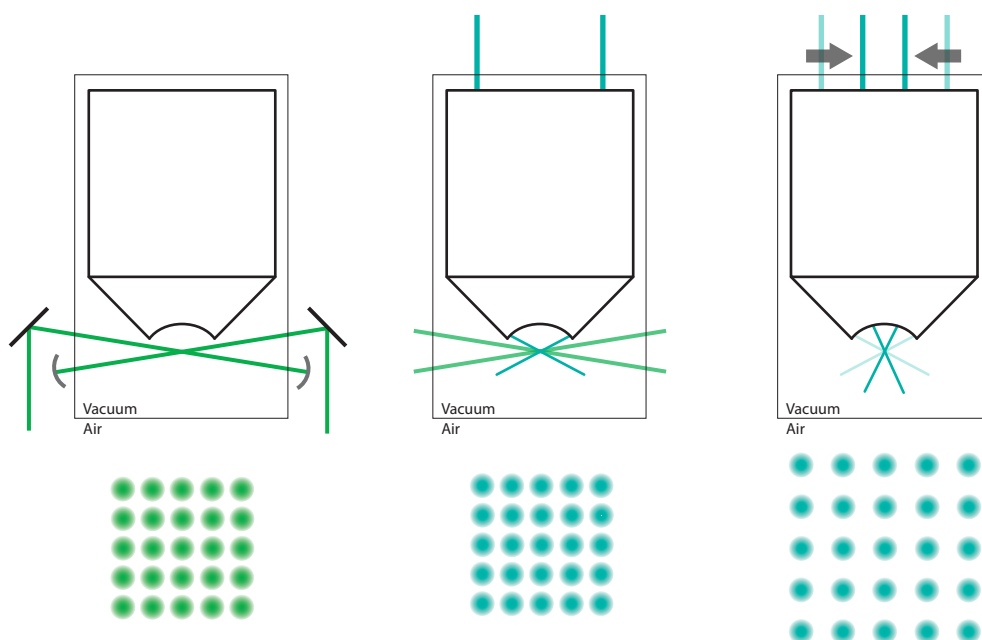


Figure 6.2: Horizontal accordion lattice sequence. After being loaded into one layer of the final vertical lattice, the atoms are loaded into the initial, low-noise, 2D horizontal lattice (left). After performing the desired science in the initial horizontal lattice, the atoms are transferred into the horizontal accordion lattice (center), whose spacing at the beginning of the sequences matches the spacing of the initial “science” lattice. The spacing of the horizontal accordion lattice is then expanded (right) to separate the atoms enough for high-fidelity imaging.

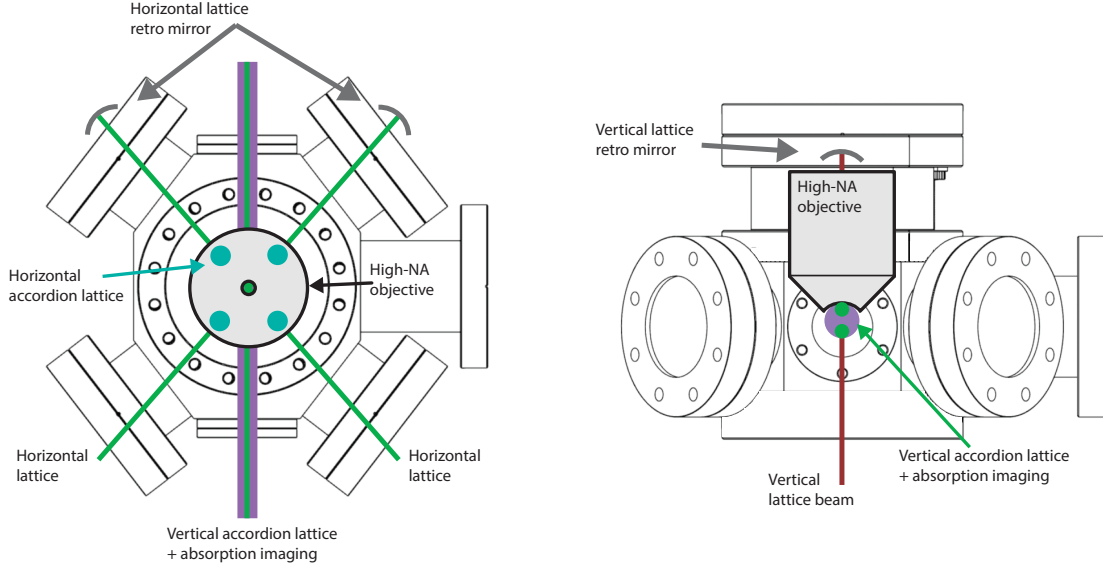


Figure 6.3: Top view (left) of the main chamber, showing all the horizontal lattice beams: the two horizontal “science” lattice beams, along with their retro mirrors, and the four horizontal accordion lattice beams. Side view (right) of the main chamber, showing all the vertical lattice beams: the two accordion lattice beams co-propagating with the low-NA absorption imaging beam, and the final vertical lattice beam, along with its retro mirror.

cal lattice is given by

$$s = \frac{\lambda}{2 \sin(\theta)}$$

Where λ is the wavelength of the laser light and θ is the half-angle between the two beams. The larger the angle between the two beams, the smaller the spacing that can be achieved. For a given wavelength, a retro-reflected lattice with $\theta = 90^\circ$ gives the smallest possible spacing of $s = \lambda/2$. Using a lens to project the lattice beams onto the atoms, we can control the angle at which the beams are focusing and interfering with each other, which in turn changes the lattice spacing as seen in Figure 6.4. For a given focal length of the lens, the separation of the two beams on the lens determines the angle at which the beams interfere after the lens. In other words, the higher the NA of the beams, the smaller the features that can be created, and the smaller the lattice spacing. By having a way of changing this separation in a continuous way, we now have a continuous control over the

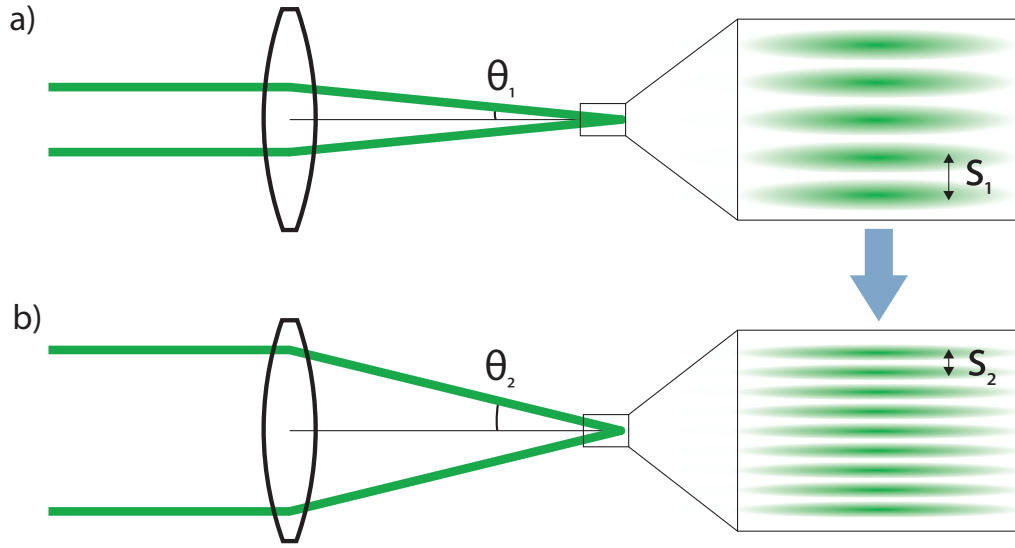


Figure 6.4: An optical lattice is created by projecting two parallel beams through a lens. The two beams intersect at the focus of the lens and interfere there, creating a lattice. The separation of these two incoming parallel beams determines the lattice spacing. In a) the incoming beams are closer together, creating a large spacing lattice, whereas in b) the separation is larger, creating a small spacing lattice.

lattice spacing on the atoms.

6.1 DESIGN OF ACCORDION LATTICES

6.1.1 DESIGN OVERVIEW

In order to truly expand the lattice around a centered, stable fringe that does not move while the lattice spacing is changed, the relative phase of the two lattice beams must be stable as the lattice spacing is changed. Any fluctuations in the relative phase of these two beams will cause a translation in the position of the optical lattice rather than only an expansion around one stable, central fringe. In order to achieve this phase stability, we use a custom beamsplitter whose design is described in detail in Section 6.1.2. In this way, all phase fluctuations that occur prior to the beamsplitter in the beam path are common-mode between both of the lattice beams and will not affect the interference

pattern. An example of such phase fluctuations is changes in the optical path length due to air currents. This custom beamsplitter is made from two dove prisms, as shown in Figure 6.7. If the two dove prisms are perfectly aligned, then a beam incident on the input face, which will split into two upon hitting the beamsplitter coating surface, will then exit the beamsplitter as two parallel beams on the output. The separation of these two parallel beams will be dependent upon the position of input beam on the input face, as can be seen in Figure 6.7. By using a lens to focus these two beams, we now have a system allowing us to control the spacing of our lattice by simply translating the position of this input beam.

One further challenge is that, when changing the lattice spacing, this beam must be truly translated across the face of the beamsplitter, with no change in angle. This could cause a change in the overlap of the two beams at the focus, changing the overall confinement. To address this, there are two potential solutions that we considered. We could either use a periscope to offset the beam, or an acousto-optic deflector (AOD).

Acousto-optic deflectors (AODs) are similar to acousto-optic modulators (AOMs) but are optimized for controlling the position of the deflected beam rather than for frequency or power control of the beam. AODs are driven with a radio-frequency signal of constant power and varying frequency which changes the deflection angle of the output beam.

A collimated Gaussian beam of wavelength λ and width D has a divergence $\Delta\Theta = \lambda/D$. The beam can be deflected by an angle $\Delta\theta$. The resolution of an AOD is measured by the number of angles that can be addressed, in units of the beam divergence angle. This number of points of resolution is equivalent to the number of spots that can be resolved once the beam is focused. Therefore an AOD has a number N of spots of resolution given by [64]

$$N = \frac{\Delta\theta}{\Delta\Theta} = \frac{\Delta\theta D}{\lambda}$$

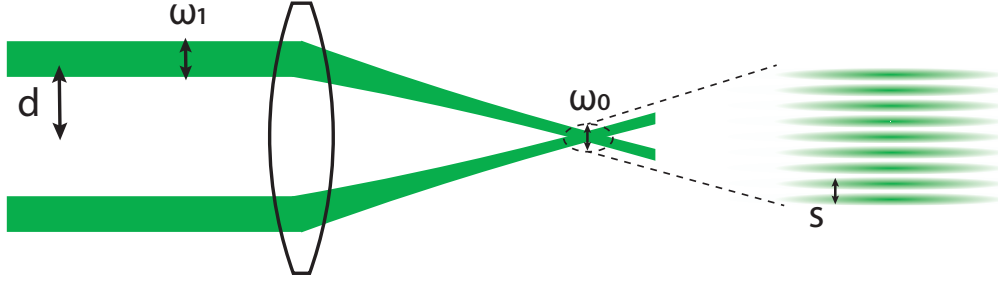


Figure 6.5: The spots of resolution required to achieve the accordion lattice is given by the ratio of the beam translation to the initial beam waist, d/ω_1 . Equivalently, this quantity can also be expressed as the ratio of the final beam waist to the smallest lattice spacing, ω_0/s . Note that this resolution is independent of the focal length of the lens, it is a property of the beam sizes and the feature sizes.

In order to reach the highest resolution of the AOD, the input beam should fill the aperture. A larger beam is less divergent, therefore reaches higher resolution within the deflection angle range $\Delta\theta$. Many AODs have apertures which are much wider than tall, due to design constraints, which means that in order to reach the maximum resolution, the beam through the AOD needs to be elliptical. Adding cylindrical telescopes before and after the AOD to change the aspect ratio of the beam would complicate the setup, so will be avoided if possible. Most typical AODs have a resolution of less than about 1000 spots, when using the full elliptical aperture.

The number of spots of resolution N_{points} required for the accordion lattice systems can be thought of in two ways. It can be found by the ratio of the maximum distance d that the incoming beam is translated on the final lens, to the waist size ω_1 of the incoming beam on the final lens, as seen in Figure 6.5. Intuitively this makes sense in the context of our accordion lattice as this measures the number of beam waists that the beam is moved by as the lattice spacing is tuned.

$$N_{\text{points}} \sim \frac{d}{\omega_1}$$

Equivalently, the points of resolution N_{points} can be thought of as the ratio of the waist size on the atoms ω_0 to the minimum lattice spacing s_{min} . Again, this is shown in Figure 6.5. Again, this

makes sense intuitively as it measures the number of times the smallest feature (the smallest lattice spacing) is repeated in the image plane. The smallest features occur when the beams are at the highest NA.

$$N_{\text{points}} \sim \frac{\omega_0}{s_{\text{min}}}$$

It is useful to note that this resolution is independent of the focal length of the lens. It is a property of the final beam size and smallest feature size. This means that we cannot get around this resolution requirement by simply changing the choice of lenses.

For the vertical accordion lattice, with a final beam waist of about 120 microns and a smallest spacing of about 2 microns (see Section 6.2 for more details), the number of spots of resolution needed is

$$N_{\text{points,vertical}} \sim \frac{120 \text{ microns}}{2 \text{ microns}} = 60 \text{ spots}$$

The spots of resolution needed for the horizontal accordion, with a final beam waist of about 100 microns and a smallest spacing of about 0.2 microns (see Section 6.3 for more details) is

$$N_{\text{points,horiz}} \sim \frac{100 \text{ microns}}{200 \text{ nm}} = 500 \text{ spots}$$

These estimates tell us that for the horizontal accordion lattice, we need more points of resolution than are typically available in AODs, if constrained to using circular beams. Since it appears that AODs don't incorporate smoothly in both the vertical and horizontal accordion systems, we use a home-built periscope that we can design to operate conveniently for both these systems. Another reason for which the periscope design is favorable for us over the AOD is that there is no power loss on the periscope (other than the small absorption on the mirrors), whereas an AOD has limited diffraction efficiency. However, AODs remain a good option for accordion lattice systems

with resolution and for a design that has more overhead of optical power. The periscope, described in detail in Section 6.1.3, is interferometrically aligned to ensure true translation of the beam.

6.1.2 BEAMSPLITTER

This custom beamsplitter is aligned and glued in-house. It is made from two off-the-shelf, NBK7 dove prisms from Thorlabs (part number PS993), which have been coated with a custom anti-reflection and beamsplitter coating as shown in Figure 6.7. The anti-reflection coating on the entrance and exit faces of the prisms is a broadband coating giving a reflection of $< 2\%$ from 360nm - 700nm light. The coating is designed for *S*-polarization and for an incidence angle of 45° , as shown in Figure 6.7 B). The beamsplitter coating is also designed for *S*-polarization, and for 360 - 700 nm light, but with an incidence angle of 72.7° . It is designed for a NBK7 to NOA61 (the epoxy we're using) interface. Ideally the coating should provide a 50/50 splitting of the incoming beam into two beams of equal optical power so as to have maximal interference contrast in the lattice.

However, the intensity of light is given by the square of the amplitude of the light wave. When two waves of amplitude A_1 and A_2 interfere constructively, the resulting peak amplitude A_{int} is given by the sum of the amplitudes of the interfering waves: $A_{\text{int}} = A_1 + A_2$. The resulting peak intensity of the interfering beams is then given by the following:

$$I_{\text{int}} \propto A_{\text{int}}^2 = (A_1 + A_2)^2 = A_1^2 + A_2^2 + 2A_1 \times A_2$$

This can be rewritten in terms of the intensities of the individual beams as:

$$I_{\text{int}} = I_1 + I_2 + 2\sqrt{I_1 \times I_2}$$

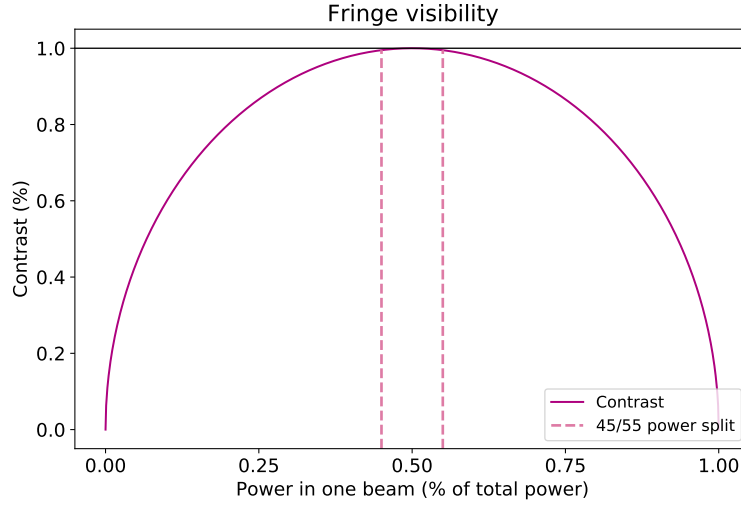


Figure 6.6: The contrast of interference fringes is plotted as a function of the splitting of the total power into the two beams. The contrast is maximal when the total power is evenly split between the two interfering beams, at 0.5. However, the contrast scales quadratically with this splitting, and if the power is imbalanced by even 45%-55%, as shown above, the interference contrast is still 99% as good.

The visibility of the fringes, or the interference contrast, is

$$\nu = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{2\sqrt{I_1 \times I_2}}{I_1 + I_2}$$

We have a constraint on the total intensity of the interfering beams $I_1 + I_2 = I_{\text{tot}}$. With this in mind, the intensity contrast of the resulting interference pattern is the following

$$\nu = \frac{2\sqrt{I_1(I_{\text{tot}} - I_1)}}{I_{\text{tot}}}$$

The quadratic scaling is in our favor here. An exact 50/50 split of the power between the two beams gives the maximum contrast, but as shown in Figure 6.6, the contrast is still 99% as high with a splitting of 45/55, for example. With this in mind, we had the prisms coated with a beamsplitter coating designed with the goal of reaching a 50/50 splitting, but with a relatively loose tolerance.

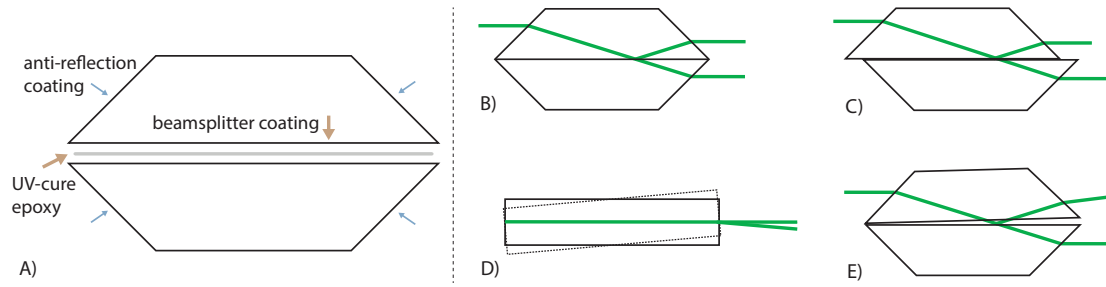


Figure 6.7: On the left, A) shows how the beamsplitters are assembled from two Thorlabs dove prisms. The anti-reflection coatings are labeled on the input and output faces of the dove prisms, and the beamsplitter coating is labeled on the face of one of each of the prism pairs. A thin layer of epoxy holds the assembly together. On the right, sketches of alignment of the two dove prisms forming the accordion lattice beamsplitter. B) shows the desired alignment, in which the two beam paths through the prism are equal and the two beams exit the beamsplitter with the same direction of propagation, they are simply offset from each other. C) shows a misalignment of the two prisms leading to unequal beam paths. D) and E) show two examples of misalignments of the two dove prisms leading to the beams exiting the beamsplitter not parallel relative to each other.

The final coating design varies by up to 2% (48/52 splitting) depending on the wavelength. For our two wavelengths of interest, the ratio is about 49.5/50.5 for 488 nm and about 50/50 for 532 nm, meaning that the interference contrast will be >99% for both our accordion lattices. Both the anti-reflection coating and the beamsplitter coating were designed and made for us by Optical Filter Source.

The two dove prisms are aligned and epoxied together using the UV cure epoxy Norland Optical Adhesive 61 purchased through Thorlabs. The desired alignment of the two prisms is such that the two beams coming out of the beamsplitter are exactly parallel, for all beam separations (in other words, for all incoming beam positions). This requirement ensures that as we tune the lattice spacing, the two beams are focused by the last lens to the same spot. A misalignment of the prism (see Figure 6.7 for examples) causing the beams to come out of the prism at an angle relative to each other would mean the beams would not get focused onto the same spot.

In addition, for the lattice to be stable as we change the spacing, we need the relative phase of the two beams to be constant as we change the spacing. This means that the difference in length

of the two optical beam paths must not change as the spacing is changed. One step further would be the requirement that the two beam paths be equal. This means not only does the relative phase between the beams remain constant as the lattice spacing is changed, but the phases are the same and remain equal as the spacing is changed. While this is not necessary for our current system, it would allow this design to be used to create optical lattices with an incoherent light source. If the beam paths are equal up to the coherence length of the light source used, the two beams will then interfere when focused together. Satisfying this stricter requirement of having equal beam paths through the beamsplitter for all lattice spacings will, by definition, give us the requirement necessary for our case, that the relative phase is stable for all lattice spacings.

With this in mind, the beamsplitters are aligned so as to meet this more stringent requirement, as it gives us a relatively straightforward method of alignment. If we simply look by eye down the beam paths through the prism at a mirror, an interferometer is formed, as shown in Figure 6.8. If the two paths are equal, up to the coherence length of the light illuminating the beamsplitter, then the light will interfere at the beamsplitter interface. Using these fringes as our signal gives us a simple, robust method of aligning the prisms.

We mount the prisms in a home-built alignment rig, shown in Figure 6.9. The prisms are each placed in a U-shaped aluminum mount and held in place with nylon screws. The bottom prism clamp is screwed on a torsion plate. The torsion plate is held to an outer frame by three thin steel rods, as shown in the figure, and is placed on a sheet of rubber a few millimeters thick. The steel rods allow the plate to tip in and out of the $x - y$ plane, with the wiggle room allowed by the sheet of rubber, but not to rotate about the z axis. The torsion plate assembly is then mounted on a rotation stage. This gives us control over the relative angle of the two prisms in the $x - y$ plane (about the z axis) but leaves the other two angles unconstrained. The top prism clamp is screwed onto an assembly of translation stages (not shown in the figure), giving us control over the relative x - y - z positioning of the prisms. The rubber under the bottom prism's torsion plate additionally ensures

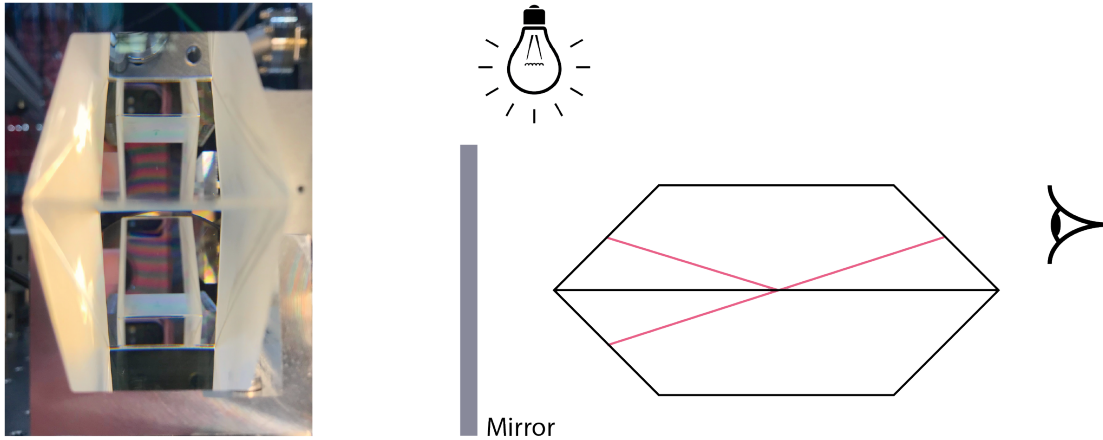


Figure 6.8: Picture (left) taken down the axis of the beamsplitter towards a mirror, as shown in the diagram (right). The interference fringes are visible in this picture, indicating that the two beam paths through the beamsplitter (as shown in diagram on the right) are equal.

that we will not accidentally apply too much force on the prism assembly with the z stage and risk scratching the interface.

During the alignment process, we first cover the ends of the beamsplitters (the entrance and exit faces) with a residue-free tape to protect from epoxy that will inevitably trickle out from between the prisms during alignment. We then wipe the two surfaces forming the beamsplitter interface and inspect them with a very bright light to ensure that no dust or marks are trapped between the prisms. We then mount the bottom prism, apply a line of the NOA61 epoxy down the middle of the surface, then carefully mount the top prism and lower it with the z stage until it touches the epoxy. We lower the prism a bit further, to compress the epoxy and push it over the entire beamsplitter surface. At this point, we wipe the excess epoxy that leaks out. We then position the prisms so that they look well-aligned, then we adjust the alignment looking for these white light fringes. For first alignment, we use a mercury vapor lamp, with a coherence length on the order of 300 microns. Once we see fringes with this source, we then switch to our “white” room light, with a coherence length estimated to be on the order of a few microns, and tweak the alignment to reproduce the

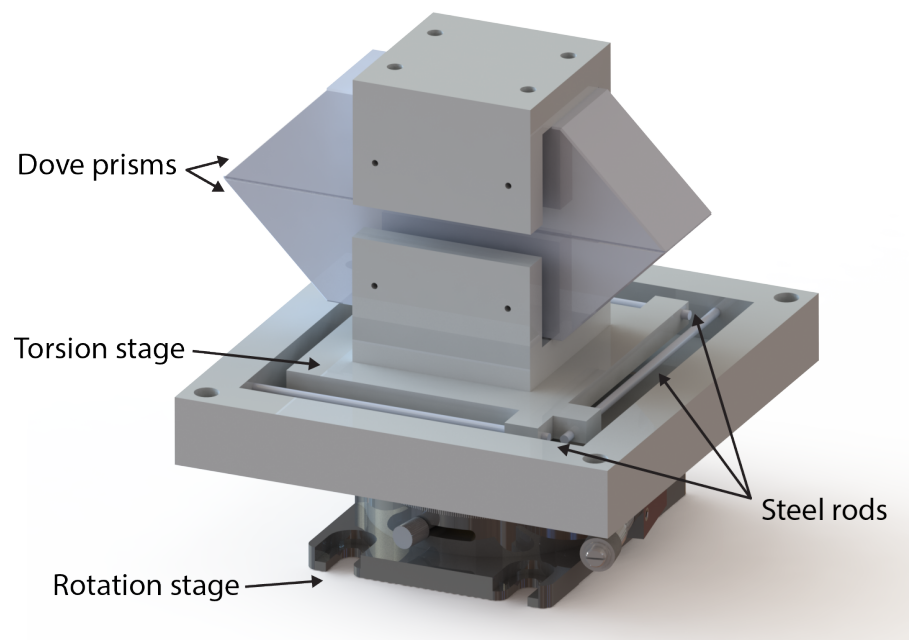


Figure 6.9: Alignment rig for the accordion lattice beamsplitters. The two dove prisms are held separately, each in a clamp. The bottom prism clamp is mounted on a torsion plate, which is secured by three steel rods to the outer frame of the mount, and is resting on a piece of rubber. The whole mount is mounted on a rotation stage. The top prism clamp is mounted to an X-Y-Z translation stage (not shown).

fringes and maximize the interference contrast.

The prisms are now aligned to match the two beam paths within a few microns. We then carefully wipe any excess epoxy and clean the input and output faces with methanol before using a high-power, 365 nm UV lamp (Thorlabs CS20K2) to cure the epoxy. We first do a quick few minutes of UV curing from all different angles and visually check that the interference fringes are still present. Then we do a longer UV cure for a few hours to ensure that the epoxy is fully set. The first prism was glued about three years ago, and there has been no measurable change or misalignment over time in the beamsplitter since.

6.1.3 PERISCOPE

The periscope, which is responsible for translating the beam up and down across the input face of the beamsplitter, is shown in Figure 6.10. This is the component that is responsible for tuning the lattice spacing. It is designed to provide 30 mm of travel, which is the height of the beamsplitter's input face. The periscope consists simply of two parallel mirrors that are 6×50 mm, and separated by 47 mm. Thus, rotating the periscope around its center changes the offset of the outgoing beam. Any deviation between the two mirrors from parallel will result in the outgoing beam being angled, not just offset. This would result in the beam having a different incidence angle on the beamsplitter as the periscope is rotated, which would result in a change of the location at which the two beams focus together.

For this reason, we align the two mirrors of the periscope interferometrically, using a simple Michelson interferometer shown in Figure 6.10. The mirrors are from broadband, dielectric, square mirrors from Thorlabs (part number: BBSQ2-Eo2), that we had them custom cut into 6 mm strips. For each periscope, we machine two thin plates between which we sandwich the mirrors. These plates are made from fiberglass, so as to add as little mass to the system as possible. We use two plates to help protect the mirrors. We glued the mirrors onto the fiberglass plates with the same UV cure

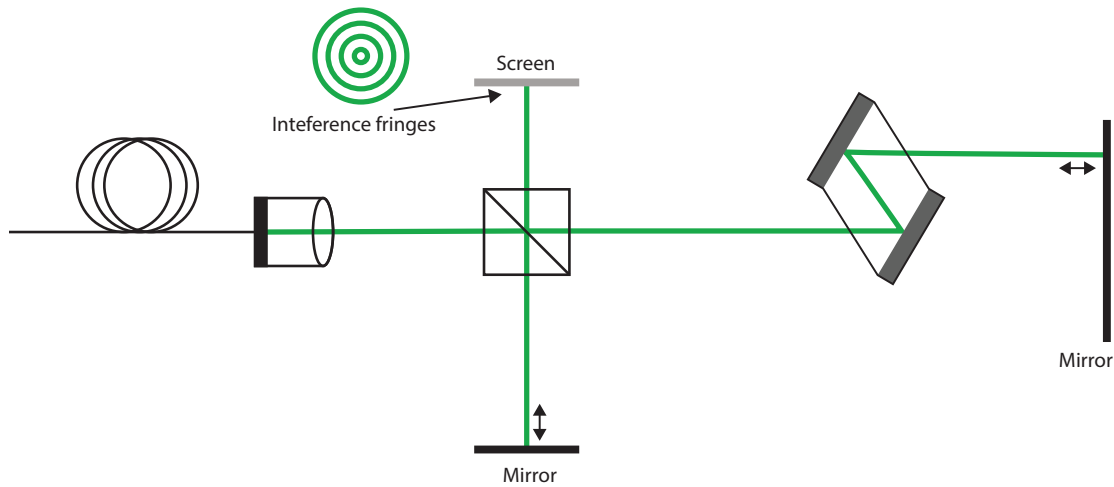


Figure 6.10: Michelson interferometer used to align the two mirrors for the periscope. The two arms of the interferometer are aligned, without the periscope in place, so as to see fringes on the screen. The periscope is then placed in one of the arms, and the relative position of the two periscope mirrors is tweaked until the fringes reappear on the screen.

epoxy Norland Optical Adhesives NOA 61. This epoxy shrinks as it cures, so in the future using an epoxy that is more stable throughout its cure might be a bit easier.

We start by aligning the Michelson interferometer without the periscope in place. We glue one of the periscope mirrors on the fiber glass plate, and mount the other mirror with double sided tape on a kinematic mirror mount. We place this kinematic mount so that the second mirror rests in the correct location on the fiber glass plate, with a layer of the UV cure epoxy. The mount gives us precise control over the mirror's angle. We place this periscope assembly in the beam path of one of the arms of the Michelson, as shown in the figure. We then use the knobs of the kinematic mirror mount to adjust the angle of the second mirror until we recover the interference pattern on the screen of the Michelson. The fringes reappearing tells us that the periscope is simply translating the beam, leaving the k -vectors unchanged. We then use a UV lamp to cure the epoxy very briefly. At this point the epoxy shrank and cause the Michelson fringes to disappear, so we tweak the mirror position a little bit before finishing the cure of the epoxy. We then glue the second fiber glass plate on top of the two already-aligned mirrors. This second plate serves mostly just to physically protect

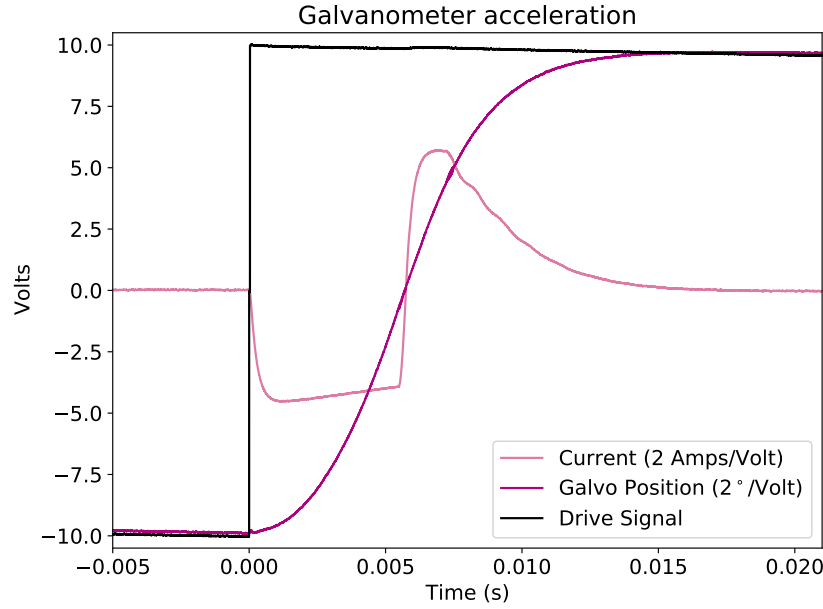


Figure 6.11: Rotation of the periscope mounted on the galvanometer after tuning the servo boards. The position of the galvo (dark pink) lags behind the drive signal (black) by about 15 ms. The current drawn from the power supply peaks at about 10 Amps, which is the maximum available in the current setup.

the mirrors.

We can quantify the precision of the alignment of the periscope mirrors by looking at the separation of the interference fringes on the screen of the Michelson interferometer. In general, the separation Δx of the fringes of two interfering plane waves of wavelength λ is given by:

$$\Delta x = \frac{\lambda}{2 \sin \alpha}$$

where α is the half-angle between the k -vectors. Both during initial alignment of the Michelson and during alignment of the periscope, we were able to easily achieve a fringe spacing of about 1 mm or greater. For the 532 nm light that we are using in this interferometer, given a fringe spacing of 1 mm, this gives an angle between the two plane waves (one from each arm of the Michelson interferometer) of about 0.02° . This number gives us an upper bound on the estimate of the angle

deviation of the beam due to the periscope.

Each mirror weighs about 4 g, giving a total moment of inertia of roughly $I = 100 \text{ g-cm}^2$. In order to rotate this large of a moment of inertia, we use galvanometers from Cambridge Technology (MicroMax Series 670). This series of galvos are designed to precisely rotate inertial loads of up to $I = 10^5 \text{ g-cm}^2$. We tune the galvo driver circuits, following the procedure in the servo board manual, with the periscopes mounted on the galvos.

The galvos have a range of $\pm 20^\circ$, driven by an input voltage with a scaling of $2^\circ/\text{Volt}$, meaning that an input drive voltage of $\pm 10\text{Volt}$ is supplied by the servo board in order to use the full range of the galvos. We drive the servo boards with a $\pm 28\text{Volt}$ supply, which gives the fastest large-angle step-response time. After tuning the response of the servo to optimize how quickly it can rotate our moment of inertia, we were able to reduce the response time to about 15 ms, as shown in Figure 6.11. This response time is limited by the current available from our power supply, so if necessary, this response could be made faster by adding a capacitor.

6.1.4 PHASE PLATE

In order to have control over the position of the maxima and minima of the accordion lattice, so that the lattice pattern can be aligned onto the atom cloud, we must have control over the relative phase of the two lattice beams. This will allow us to move the central, stable fringe that doesn't move as the lattice spacing is changed and align it onto the atoms. This control over the relative phase of the two lattice beams can be achieved by placing a glass plate in the path of one of the beams. Rotating this plate changes the beam's angle of incidence on the plate, which changes the beam's path length through this higher-index material. The angle of the plate now gives us a way of tuning the path length difference between the two beams, and thus shifts the intensity pattern.

The presence of this plate introduces a significant relative phase shift in between the two beams. For a light source with a long coherence length, this is not an issue as the two beams will continue to

interfere. We simply want to have a way of tuning to position of the interference pattern. However, if one were to use a similar system to generate an accordion lattice with an incoherent light source, the shift from adding a plate in one of the beam paths would be enough to prevent the beams from interfering. A workaround for this would be to place a glass plate of the same thickness in the path of both lattice beams, and only rotating one to shift the relative phase.

We use a simple off-the-shelf plate with an anti-reflection coating. It is rotated by a Thorlabs galvanometer. There is no need for a more powerful galvanometer, since this is a small, light optic.

6.2 VERTICAL ACCORDION

For the vertical accordion lattice, the initial parameters need to be optimized on the constraint of capturing as much of the initial quantum gas as possible. To achieve maximal capture, the initial spacing of the lattice needs to be larger than the size of the atom cloud, and the initial lattice has to be deep enough to support the atoms against gravity. The initial spacing should be large enough to fully capture the degenerate cloud of atoms from the ODT, which has a vertical height of $20\text{ }\mu\text{m}$. The final accordion lattice parameters need to be optimized so as to load all the atoms into one plane of the vertical lattice. Any atom that is accidentally loaded into the adjacent planes of the vertical lattice will be out of the focal plane of the objective and will cause background noise in the images. To achieve the highest transfer efficiency possible, the relevant parameter is the trap frequency of the final accordion lattice. This trap frequency determines how strongly the atoms are confined, and determines the spatial extent of the wavefunction. For optimal transfer, the wavefunction should be completely confined to one lattice spacing of the final vertical lattice we are transferring to. Our final vertical lattice is a retro-reflected 1064 nm , with a spacing of 532 nm . Therefore we need the final accordion lattice trap frequency to be such that the wavefunction of the atoms is confined within $\pm \frac{532}{2}\text{ nm}$.

We can approximate the wavefunction of the atoms in the lattice as the ground state of a harmonic oscillator, with $\alpha = 2\pi\omega_{\text{trap}}/\hbar$ and ω_{trap} is the axial trap frequency:

$$\Psi_{HO}(x, \omega) = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2}$$

The fraction F_{transfer} of the wavefunction contained within one lattice site of the final static vertical lattice with spacing $s_{\text{final}} = 1064 \text{ nm}/2 = 532 \text{ nm}$ is then given by the following integral:

$$F_{\text{transfer}} = \frac{1}{N} \int_{-s_{\text{final}}/2}^{s_{\text{final}}/2} \Psi_{HO}(x, \omega_{\text{trap}}) \times \Psi_{HO}^\dagger(x, \omega_{\text{trap}}) dx$$

As a sanity check we can verify numerically that $N = 1$, the normalization given by the integral:

$$N = \int_{-\infty}^{\infty} \Psi_{HO}(x, \omega_{\text{trap}}) \times \Psi_{HO}^\dagger(x, \omega_{\text{trap}}) dx = 1$$

The transfer efficiency's dependence on the trap frequency is shown in Figure 6.12. A transfer efficiency of more than 99.9% requires an axial trap frequency of more than 4.6 kHz.

In order to achieve the relatively high required final trap frequency with our available powers, it is helpful to have the final spacing of the lattice be as small as possible. For interfering beams at an angle θ , the axial trap frequency is affected by the angle. The radial trap frequencies is independent of the angle of the beams. The resulting trap frequencies are given by:

$$\omega_{\text{trap,xy}} = \sqrt{\frac{4}{m \times \text{waist}^2} \times \alpha_{\text{pol.}} \times I},$$

$$\omega_{\text{trap,z}} = \frac{1}{2\pi} \sqrt{\frac{4\pi\hbar}{m} \times \frac{2\pi \sin \theta}{\lambda} \times \alpha_{\text{pol.}} \times I}$$

where m is the mass of the atom, $\alpha_{\text{pol.}}$ is the polarizability of erbium at the laser wavelength λ ,

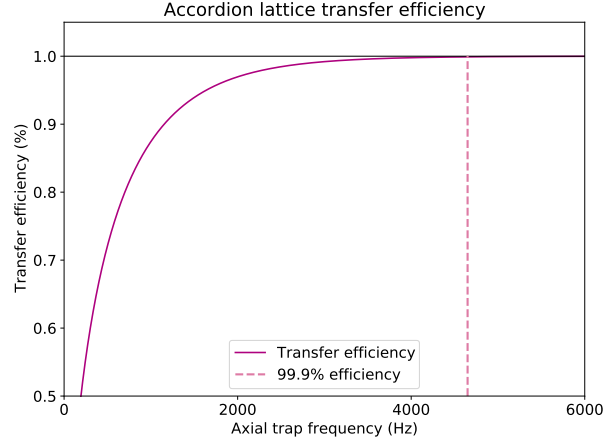


Figure 6.12: Estimated transfer efficiency from the accordion lattice to the final vertical lattice. To obtain an efficiency greater than 99.9%, an axial trap frequency of greater than 4.6 kHz is needed.

and I is the intensity of the lattice beams. We are using light from the same system that generates light for the 2D horizontal lattice. This is 532 nm light produced by single-pass doubling of light from a 1064 nm Azur Light Systems fiber laser. After accounting for the power necessary for the horizontal lattice, this system is providing us with approximately 1 Watt of additional 532 nm power for the vertical accordion lattice. With this power limitation, and using the maximum NA possible restricted by the viewports which is $NA = 0.08$, the absolute maximum trap frequency attainable is about 4 kHz. This gives a transfer efficiency of about 99.7%.

For that reason, we place the final lens projecting the beams onto the atoms inside the vacuum chamber, as shown in Figure 6.17. This increases the maximum NA to $NA = 0.14$. This increase actually gives a large enough boost in the trap frequency to significantly improve the transfer efficiency. The maximum achievable trap frequency at this NA is about 7 kHz, giving a transfer efficiency of 99.99% into a single pancake. Projecting the accordion lattice beams at this NA gives a spacing of 1.9 microns.

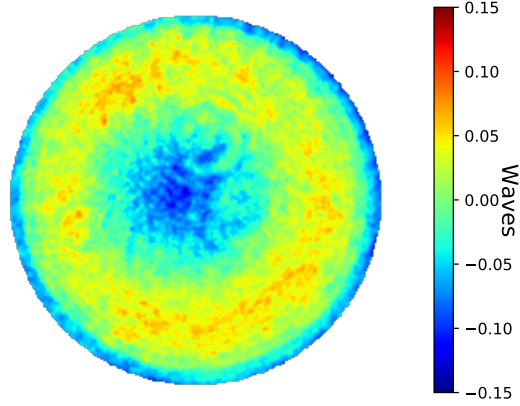


Figure 6.13: Phase map of the full clear aperture of the aspheric lens, with the tilt and focus aberrations removed.

6.2.1 IN-VACUUM ASPHERIC LENS

The lens placed inside the vacuum chamber to project the vertical accordion lattice beams is a custom, in-house aspheric lens design. The size and focal length of this lens is constrained by the geometry of the vacuum chamber, meaning no off-the-shelf option was suitable. Taking advantage of the flexibility offered by already having a custom lens, we also designed this lens for absorption imaging of the atoms with 401 nm light.

For the accordion lattice application, the lens must simply focus down small beams of 532 nm light to a specific focus point. Since the lattice beams are small, there is no concern of aberrations on the lattice beams caused by imperfections in the lens surfaces. For the absorption imaging application, we want diffraction-limited performance at 401 nm. The lens will have a focal shift between the 532 nm and the 401 nm light, meaning only one of the colors will be focused on the atoms by

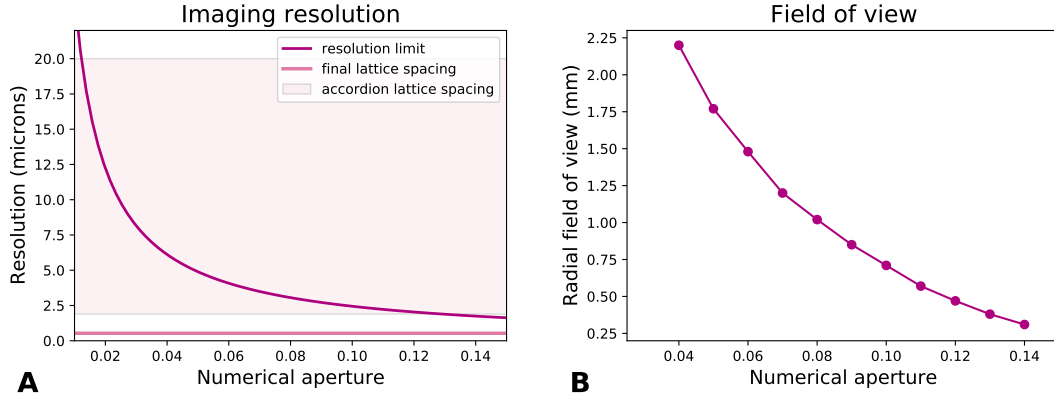


Figure 6.14: A) shows the scaling of resolution of the absorption imaging on the numerical aperture. The final vertical lattice spacing is too small to be resolvable given the range of numerical aperture available. However, much of the range of lattice spacing of the vertical accordion lattice is resolvable. B) shows the simulated (in Zemax) scaling of the field of view of the aspheric lens with the numerical aperture.

this lens. Luckily, it is easy to compensate for a defocus in the imaging light with additional lenses outside of the chamber, so the lens is designed to be placed such that the 532 nm light is focused on the atoms.

The lens is a single aspheric surface, made by Optimax from Corning HP Fused Silica 79-80 with an anti-reflection coating from 400-600 nm. The lens was designed using the optical design software Zemax, the version available to us in the lab. The final aspheric surface parameters are shown in Table 6.1.

The aspheric surface of the lens is optimized up to $NA = 0.14$ at 401 nm, which gives us the flexibility of using the lens for imaging at any numerical aperture up to $NA = 0.14$. Given a finalized lens surface, there is then a trade-off between the desired field of view and the numerical aperture up to which the lens is used, as shown in Figure 6.14 B). This translates to a trade-off between ease of alignment and imaging resolution. The atom location is determined by the high-resolution objective, so this aspheric lens must be placed in the chamber such that the atoms are within its field of view for imaging. The higher the NA we desire, the more precisely we must be able to place this lens

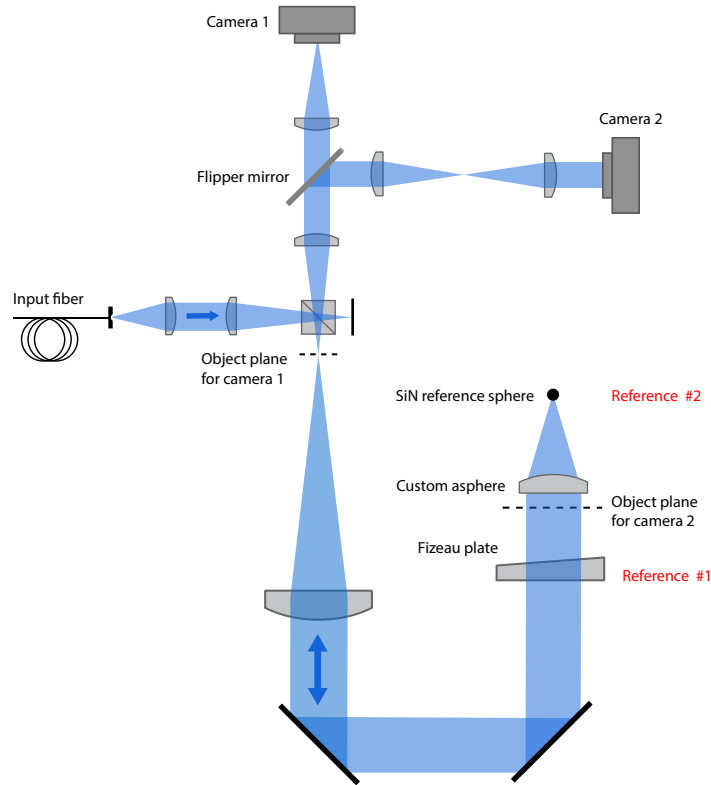


Figure 6.15: Fizeau interferometer used to measure aberrations from the custom in-vacuum aspheric lens for the vertical accordion lattice. Figure adapted from Aaron Krahn [30].

relative to the objective, to make sure the atoms are still within the field of view. If the lens is slightly misplaced so that the field of view must be extended, this means the NA must be reduced.

The resolution of an imaging system is given by

$$\text{resolution} = \frac{1.22\lambda}{2 \times \text{NA}}$$

The interesting length scales for us to image are the pancakes of the final vertical lattice (made by retro-reflecting 1064 nm light) which are separated by 532 nm, and the pancakes of the vertical accordion lattice, which have a separation in the range of about 2-20 μm . Given the geometry of

Aspheric lens parameters	
Glass type	HPFS C79-80
Center thickness	8.00 mm
Conic	-0.613838
Radius	-23.72908 mm
R ₄	-2.960199×10^{-7}

Table 6.1: In-vacuum aspheric lens design parameters, obtained from an optimization run with the Zemax Optics Studio software.

the system we will never be able to have high enough NA to image the individual layers of the final vertical lattice, but the layers of the accordion lattice are within reach for imaging NA of less than $\text{NA} = 0.14$. This is summarized in the left plot of Figure 6.14. We aim for an NA of about 0.1-0.08 which gives a field of view of about 1 mm, as seen in the right plot of Figure 6.14. For this reason, we are able to relax the alignment criteria of the aspheric lens relative to the objective to a level that can be reached by carefully designing and referencing the mounting of the objective and the lens.

We made an interferometric measurement of the aberrations of this lens. We did this using a Fizeau interferometer, which we had already set up to characterize our high-NA objective from Special Optics. The resulting phase map is shown in Figure 6.13. The phase map shows us that the peak-to-valley aberration is less than ± 0.15 waves.

The Fizeau interferometer allows us to measure the aberrations on an optic of choice (here the custom asphere). A first reference surface (labeled reference #1 on the figure) causes a plane wave to reflect back. The rest of the beam propagates through the custom asphere, is focused onto a spherical reference sphere (labeled reference #2 on the figure) and reflects back. The two plane waves, reflected from reference #1 and from reference #2 (which has propagated twice through the custom asphere) are interfered. All aberrations that are common mode between the two beams will not show up in the interference pattern. Any phase shifts in the second beam, due to aberrations in the custom asphere, will show up on the interference pattern, thus giving us a measurement of the aber-

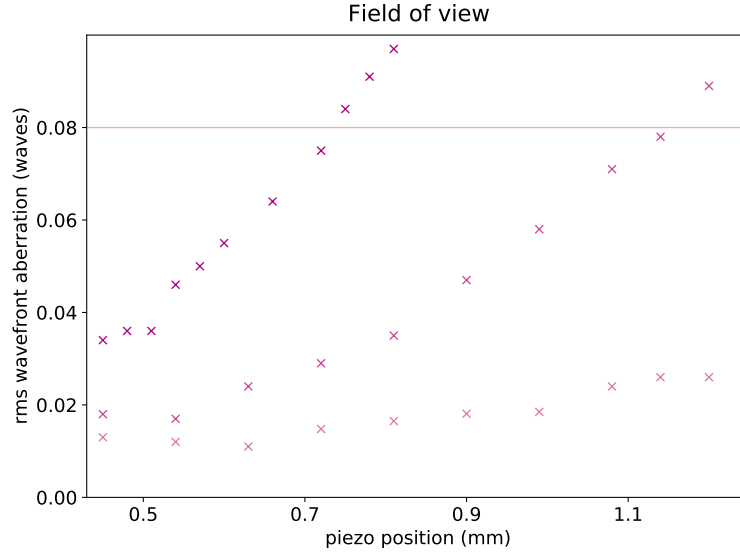


Figure 6.16: RMS wavefront aberration of the in-vacuum aspheric lens, measured with a Fizeau interferometer. Horizontal line indicates the diffraction limit over 0.08 waves of aberration.

rations. The reference #1 is on a piezo-actuated mount, allowing small displacement in the plate to explore the phase difference between the beams. More details on aberrations measurements and the Fizeau setup and its alignment are in [30]. The images were taken and analyzed with the software Intellwave.

The RMS aberration was measured for different numerical apertures, shown in Figure 6.14. The measurements made at higher NA have a much smaller field-of-view, as expected. The measurements made at lower NA are simply measurements made with a circular mask reducing the aperture of the measurement.

6.2.2 OPTICS LAYOUT AND EXPANSION IMAGES

The optics layout for the vertical accordion lattice is shown in Figure 6.17. The dichroic mirror is used to separate the accordion lattice beam path from the absorption imaging 401 nm beam path. The periscope and the phase plate, labeled on the diagram, are both mounted on and rotated by gal-

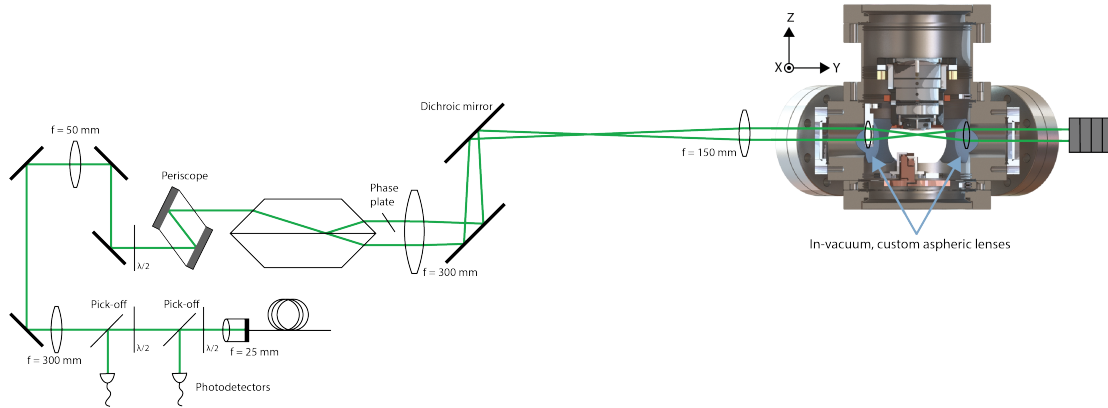


Figure 6.17: Optics layout for the vertical accordion lattice system. Both the periscope and the phase plate labeled on the diagram are rotated by galvanometers.

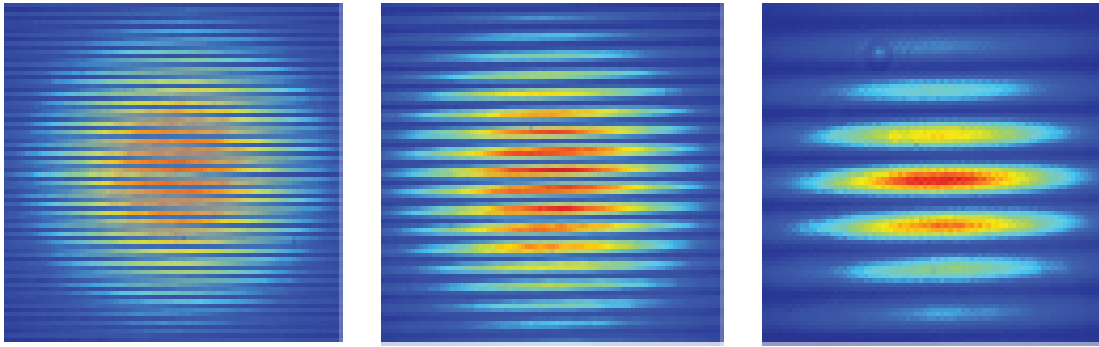


Figure 6.18: Images of the lattice spacing expanding around a central fringe. These pictures are of the light only, no atoms!

vanometers, as discussed above. The beam is first aligned through the beamsplitter at a fixed height, ensuring the two output beams come out parallel. The periscope and the first focusing lens after the beamsplitter are then added, and we verify that the beams focus in the correct spot for all separations (that is, for all angles of the periscope). Once the lattice is forming and expanding properly, the only thing left is to image it onto the atoms with the $4f$ lenses. One thing to be careful about is to not twist the lattice pattern with the alignment mirrors after the prism beamsplitter.

Resulting fringes are shown in Figure 6.17. We measured a drift of the central fringe of less than 0.3 waves over an expansion by a factor of 10 of the spacing.

6.3 HORIZONTAL, 2-DIMENSIONAL ACCORDION LATTICE

The 2-dimensional, horizontal accordion lattice is used to expand the atoms at the end of the sequence, before the high-resolution imaging. The initial spacing of the horizontal accordion therefore needs to match the spacing of the science lattice, which is retro-reflected 532 nm light.

The concept of the horizontal accordion is similar to the vertical accordion. However, it is being projected onto the atoms through the high-resolution imaging objective, with an NA of 0.92. In order to match the spacing of the science lattice, this accordion therefore needs to be from a laser with slightly shorter wavelength than the science lattice laser, since the accordion beams are not exactly counter-propagating.

With a 488 nm laser projected through the maximum NA of the objective, we can achieve a lattice spacing of

$$d = \frac{\lambda_{488}}{2 \times NA} = 0.265 \text{ microns}$$

This allows us to match perfectly with the science lattice, which has a spacing of $532 \text{ nm}/2 = 0.266 \text{ microns}$. From this small initial spacing, the horizontal accordion lattice is then expanded to pull the atoms apart before the high-resolution imaging.

The design of the horizontal accordion lattice is very similar to that of the vertical accordion lattice. It includes all of the components discussed in detail in the first half of this chapter. The main differences are that the horizontal lattice is, obviously, 2-dimensional, and is being expanded along both axes. This lattice therefore is made up of two identical pairs of expandable lattice beams, each generated by a rotating periscope and a dove prism beamsplitter. The two beams are combined with a beamsplitter. The second main difference is that the horizontal accordion's lattice spacing is designed to expand by more than a factor of 10x, starting from about 260 nm and going up to 5-10 μm .

The light for this lattice is from a 2 W fiber laser from Azurlight Systems. The light is distributed into 2 photonic crystal fibers, one for each lattice axis. The AOMs controlling the power in each fiber operate at slightly different frequencies, so as to avoid interference between the light in the different axes. The fibers are then routed to the main chamber, delivering the light to the accordion setup, which is conceptually similar to the vertical accordion setup in Figure 6.17. Since the focal length of the objective is so short ($f_{eff} \approx 20$ mm) the beam sizes and lenses are different between the two setups.

7

Conclusion and outlook

RAPID BEC AND DFG PRODUCTION: With this apparatus, we have demonstrated sub-second production of nearly pure erbium Bose-Einstein condensates of 80,000 atoms. This is a promising step towards creating a dipolar quantum gas microscope with a rapid repetition rate, comparable to that of Rydberg arrays. This apparatus also allows the flexibility of easily switching between all isotopes of erbium, including a fermionic isotope. The sample preparation time for a degenerate

Fermi gas is shown here to be about 5 s, which is also an improvement on the standard for the field. The methods used in the experiment could in principle be applied to speed up the degenerate gas sample preparation for a broad range of elements.

The possibility of increasing repetition rates will allow for experiments requiring many measurements in order to sample from the wavefunction of quantum many-body systems. Measuring higher-order correlations (typically measuring above 2nd order correlations has been difficult due to limited size of the data sets) provides information about the structures and interactions in these complex systems, as in [65]. Increasing the possible sizes of data sets could also open doors to studying non-equilibrium dynamics. Other applications which would benefit from large data sets include using machine learning to classify and extract information from images [66].

ACCORDION LATTICE DESIGN: This thesis also details the design and implementation of two dynamically-tunable accordion optical lattices. In this experiment, these systems have been designed for compression of the bulk atomic cloud in one dimension, and for pinning the atoms and expanding the lattice of individual atoms in two dimensions, before imaging. However, the design of these expandable lattices could be used for creating an expandable lattice of incoherent, “white” light, or for a system in which control over the spacing of the lattice during interactions is desired.

EXPERIMENTAL DETAILS: I have also detailed many aspects of the apparatus. Details on the implementation of the two-stage magneto-optical traps, done for the first time on erbium, are given here. I outline both the experimental steps and show data on the atom numbers, the transfer between the MOTs, the lifetime of both MOTs, and the temperature behavior of the narrow, 8kHz line MOT. I also give details on the magnetic field control in the apparatus, as well as descriptions of the assembly process for our main magnetic field coil assembly. The hope is that this helps future students on the experiment understand parts of the apparatus that will get buried deep underneath the rest.

OUTLOOK: This apparatus is uniquely positioned to perform site-resolved studies of the extended Bose-Hubbard model. Studies in bulk of the extended Bose-Hubbard model have been realized with erbium in optical lattice [24], showing signatures of the dipole-dipole interaction in erbium and its effect on the system's phases. It is predicted that, for bosons in an optical lattice, the dipole-dipole interaction leads to interesting insulating metastable states [67], modifying the traditional superfluid-to Mott insulator phase diagram. A signature of these novel phases is the filling pattern of the atoms in the lattice, which a quantum gas microscope is ideally positioned to measure.

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